Virtual Ryanair Boeing 737-8AS(WL) ASOP

Aircraft Standard Operational Procedures

Prepared By:.....Signature
Ian Livesey
Virtual Ryanair Operations Manager

Approved By:.....Signature
Anna Győrffy
Virtual Ryanair Operations Manager

Issued By: .....Signature
Jack Wilkinson
Virtual Ryanair Operations Director

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Table of Contents

1. Administration and Control 5
1.1. Revision Record 5
1.2. Revision Highlights 5
2. Introduction 6
2.1. Introduction 8
2.1.1. General 8
2.1.2. Controls and Indications – Nomenclature 8
2.1.3. Autopilot Flight Director System and Flight Management System Monitoring 8...
2.1.4. FMC/CDU Operation 8
3. Preflight Procedures 9
3.1. Preliminary Flight Deck Procedure 9
3.2. Preflight Procedure 13
3.3. Final CDU Pre-flight Procedure 23
3.3.1. Final CDU Pre-flight Procedure – Guidance 24
3.4. Initial Emergency Briefing 25
3.5. Final Flight Deck Preparation 26
3.6. Before Taxi Procedure 27
4. Taxi and Takeoff 29
4.1. Taxi Procedures 29
4.2. Before Takeoff Procedure 30
4.3. Takeoff Procedure 32
4.3.1. Takeoff Roll and Rotation 33
5. Climb 36
5.1. After Takeoff Procedure 36
5.1.1. After Takeoff Procedure - NADP 1 37
5.1.2. After Takeoff Procedure - NADP 2 38
5.2. Takeoff Flap Retraction Speed Schedule 39
5.3. Altimetry Setting Procedures 40
5.4. Climb and Cruise Procedure 41
5.5. FL100 and "10 Checks" 42
6. Cruise 43
6.1. R/T Discipline 43

6.2. Climb and Cruise Procedures 43
6.3. cruise procedures 44
6.4. Immediate Level off Procedure 44
6.5. Fuel Procedures 45
6.6. Fuel Crossfeeding Procedure 45
6.7. Transponder Use 45
7. Descent 46
7.1. Initial Descent Procedures (TOD – FL150) 46
7.2. Descent Procedures FL150 and Below 48
7.3. Below FL100 49
8. Approach 50
8.1. Approach Procedure 50
8.2. Stabilised Approaches 51
8.3. The Landing Gate 51
8.4. Flap Policy 52
8.5. Flap Extension Schedule 53
8.6. Autobrake, Reverse Thrust and Flap 54
8.7. Visual Approach 55
8.8. Timed Approaches 57
8.9. Surveillance Radar Approaches, SRA 58
8.10. 500 Continue/500 Go-Around Call 59
8.11. Landing Procedure Autoland 60
8.12. Landing procedure single chanel or manual 62
8.12.1. ILS Approach – Guidance 63
8.13. Low Visibility Approaches - Guidance 64
8.14. Non precision Approaches (NPA) 65
8.15. Intrument Approaches using vnav 66
8.16. Non-Precision Approaches using VNAV 68
8.17. Circle to Land Approach 70
8.18. Non-Precision Approach using VS 72
9. GO-AROUND 74
9.1. Go-Around Procedure Dual Channel 75
9.2. Go-Around Procedure Single Channel or Manual 76

9.3. Go-Around Guidance 77
10. Landing 78
10.1. Landing Rollout Procedure 78
10.2. Runway Visual and Landing 80
10.3. Crosswind Landings 82
11. Post Flight Procedures 84
11.1. Taxi In Procedure 84
11.2. Shutdown Procedure 85
11.3. Transit Shutdown Procedure 86
11.4. APU Use 87

1. ADMINISTRATION AND CONTROL

1.1. REVISION RECORD

Issue Number	Revision Number	Date Issued	Issue Number	Revision number	Date Issued
1	**0**	28 November 2014			
		19 November 2019			
1	**1**				

1.2. REVISION HIGHLIGHTS

This is a complete re-issue of the SOP Manual with a new chapter structure that follows phase of flight sequence.

2. INTRODUCTION

The existence of a set of Virtual Ryanair SOPs is essential to the continued safe operation of our virtual fleet. Our SOPs try to provide a common language for individuals who opt for 'as real as it gets' approach in Flight Simulation. They describe the skills and knowledge required to complete a normal procedure, briefing or manoeuvre. SOPs should not be so complex that they are difficult to follow. Equally, they must provide a means for the Virtual Ryanair to ensure that regulations are complied with and repetitive errors are eradicated.

Each SOP, together with its associated documents, has been developed in response to prompt and accurate feedback from our Virtual Pilots, Vatsim ATC Controllers and Real World Pilots.

This manual is a combination of the Boeing FCOM Volume 1 content and our own Standard Operating Procedures.

The text may be displayed under 3 different headings as follows:

1. **"Section Title"**

   Text in a section titled such as this is identical to the equivalent section in FCOM Vol 1 available with PMDG Being 737. The text states the core task and procedure that relates to that task.

   The SOP section will relate to the core task and procedure relevant to the section and amplified with additional text designed to instruct the pilot more specifically through the task. The SOP will specify how the relevant procedure, briefing or manoeuvre must be completed in order to complete the task specified in the Section.

2. ***Airmanship***

   (The Airmanship Section will be written in Italics to give it a distinct look in the SOP document.)

   Airmanship has been described as "Aviation Common Sense". It is a concept which combines a pilot's professionalism, aviation knowledge, maturity and self-awareness in its definition. It is the bedrock of a safe and efficient pilot.

   Standard Operating Procedures exist to enhance airmanship not to suppress it. SOPs promote situational awareness and highlight threats, errors or omissions. If ever there is a conflict between the SOP and good Airmanship then the pilot must resolve that conflict using good judgment. Once any such conflict is resolved the default position should always be to revert to SOP.

   Good aviation common sense – Airmanship – will assist the pilot in knowing what the safe, professional and common sense thing to do is in all operations and procedures. We believe that these SOPs, developed over years and reflecting past experience, provide a Virtual Ryanair pilot with a readymade safe, professional and common sense means of operating the aircraft.

The information in these sections is intended to:

- *Anticipate threats associated with a particular phase of flight or procedure.*
- *To inform the pilot of matters related to the topic in a general sense. It will provide an industry or company background or perspective to the related material.*
- *To present material that is not an SOP but is related to the Procedure. This allows the SOP itself to be less complex.*

Not all sections of the SOP Manual will have an Airmanship entry.

3. **Guidance**

The Guidance Section will contain the Virtual Ryanair preferred means of completing the SOP. A pilot who uses the Guidance diligently will comply with the SOP and achieve its objective. It will contain proven, compliant, professional and efficient means of completing the SOP.

Not all sections of the SOP Manual will have a Guidance entry. Guidance is shown in text

2.1. INTRODUCTION

2.1.1. GENERAL

This chapter contains Normal Procedures.

2.1.2. CONTROLS AND INDICATIONS – NOMENCLATURE

Controls and indications appear in all UPPERCASE type to correspond to the words on the control panel or display. For example, the following item has UPPERCASE words to match what is found on the panel:

EQUIPMENT COOLING switches ..... NORMAL

The word EQUIPMENT is spelled out, even though it is abbreviated on the panel.

The following appears in all lower case because there are no words identifying the panel name.

Engine display control panel ..... Set

2.1.3. AUTOPILOT FLIGHT DIRECTOR SYSTEM AND FLIGHT MANAGEMENT SYSTEM MONITORING

When the autopilot, flight director, or autothrottles are in use and a MCP mode change is selected, confirmation of the intended selection must be verified only by reference to the flight mode annunciation display. Airplane course, vertical path, thrust and speed must always be monitored.

MCP altitude changes following an ATC instruction:

(1) ATC given instruction to change ALT/Level
(2) Pilot resets MCP altitude
(3) Pilot responds to ATC

2.1.4. FMC/CDU OPERATION

In flight, CDU entries should be accomplished prior to high workload periods such as departure, arrival, or holding. During high workload periods, using the autopilot modes such as heading select, level change, and the altitude and speed intervention features may be more efficient than entering complex route modifications into the CDU

3. PREFLIGHT PROCEDURES

Good Airmanship dictates that a pilot will complete this procedure carefully and diligently. The aircraft is potentially dangerous to any crewmember, service provider or engineer until this procedure has been completed. Many lives, limbs and licenses have been lost in and around an aircraft that had not been made Safe by a checklist and procedure such as this.

3.1. PRELIMINARY FLIGHT DECK PROCEDURE

Perform the following checks prior to assuming normal crew positions.

BATTERY switch ..... ON

Guard – Down

ELECTRIC HYDRAULIC PUMP switches ..... OFF

LANDING GEAR lever ..... DN

Six green landing gear indicator lights – Illuminated

GROUND POWER switch (if ground power is available) ..... ON

SOURCE OFF lights – Extinguished

Fault/Inop detection ..... Check

OVERHEAT DETECTOR switches – NORMAL

TEST switch – Hold to FAULT/INOP

Verify MASTER CAUTION, OVHT/DET annunciator, FAULT and APU DET INOP lights are illuminated.

If the FAULT light fails to illuminate, the fault monitoring system is inoperative.

If APU DET INOP light fails to illuminate, do not operate APU.

Fire/Overheat warning ..... Check

TEST switch – Hold to OVHT/FIRE

Verify fire warning bell sounds, master FIRE WARN lights, MASTER CAUTION lights and OVHT/DET annunciator illuminate.

Master FIRE WARNING light – Push

Verify master FIRE WARN lights and fire warning bell cancel.

Verify engine No. 1, APU, and engine No. 2 fire warning switch and engine No. 1 and engine No. 2 OVERHEAT lights are illuminated. If AC busses are powered, verify WHEEL WELL fire warning light is illuminated.

If an engine fire warning switch and an ENG OVERHEAT light do not illuminate, a detection loop is inoperative.

EXTINGUISHER TEST switch ..... Check

Position TEST Switch to 1, verify the green extinguisher test lights are illuminated. Release switch and verify the lights are extinguished.

Repeat for test position 2.

APU ..... Start and on busses

Note: To conserve fuel and reduce emissions, start the APU when the loadsheet arrives at the airplane unless environmental conditions dictate otherwise.

Note: If extended APU operation is needed on the ground, position an AC operated fuel pump ON. If fuel is loaded in the center tank, position the left center tank fuel pump switch ON to prevent a fuel imbalance before takeoff.

CAUTION: Center tank fuel pump switches should be positioned ON only if the fuel quantity in the center tank exceeds 453 kgs.

Note: Whenever the APU is operating and AC electrical power is on the airplane busses, operate at least one fuel boost pump to supply fuel under pressure to the APU to extend the service life of the APU fuel control unit.

When the APU GEN OFF BUS light is illuminated:

APU GENERATOR bus switches – ON

Verify that the SOURCE OFF lights are extinguished.

Verify that the TRANSFER BUS OFF lights are extinguished.

EMERGENCY EXIT lights switch ..... ARMED

Guard – Down

NOT ARMED light – Extinguished.

Cabin attendant call chime ..... Test

FLAP lever ..... Set

Position the FLAP lever to agree with the FLAPS position indicator.

Takeoff configuration warning ..... Check

Advance thrust levers to forward stop to ensure takeoff configuration warning horn sounds. Retard thrust levers to idle stop.

CARGO FIRE system ..... Check

DETECTOR SELECT switches – NORM

TEST switch – Push

Verify master FIRE WARN lights and fire warning bell cancel.

Verify cargo fire (FWD, AFT) warning lights are illuminated.

Verify DETECTOR FAULT light remains extinguished.

Verify the green EXTINGUISHER test lights are illuminated.

Verify the cargo fire bottle DISCHARGE light is illuminated.

VOICE RECORDER switch ..... ON

Verify that this switch is on, including during all turnarounds, to record all clearances, briefings and checklists.

Flight recorder ..... Test

CAUTION: This switch should not be confused with the Passenger Oxygen Switch, as this will cause deployment of the Passenger Oxygen System.

FLIGHT RECORDER OFF light – Illuminated

FLIGHT RECORDER test switch – TEST

FLIGHT RECORDER OFF light – Extinguished

FLIGHT RECORDER test switch – NORMAL

STALL WARNING TEST switches ..... Push and hold, one at a time

Verify that each control column vibrates when the respective switch is pushed.

MACH AIRSPEED WARNING TEST switches ..... Push

Verify clacker sounds.

REVERSER lights ..... Extinguished

EEC switches ..... ON

ALTERNATE lights – Extinguished

PASSENGER OXYGEN switch ..... NORMAL

Guard – Down

PASS OXY ON light – Extinguished

CAUTION: Switch activation will cause deployment of passenger oxygen masks.

CREW OXYGEN pressure indicator ..... Check

SERVICE INTERPHONE switch ..... As required

GPS light ..... Extinguished

IRS mode selectors ..... NAV

Note: Prior to commencing the alignment procedure the airplane must be parked and not moved until alignment is complete and the ALIGN lights extinguish.

Verify both ON DC lights illuminate momentarily followed by steady illumination of the ALIGN lights. The ALIGN lights will remain illuminated until the IRS enters the NAV mode.

Crews are required to perform a full realignment only for the first flight of each flight duty period. There is no requirement, under normal circumstances to realign the IRS's during subsequent turnarounds.

The UNABLE REQD NAV PERF-RNP message may show until IRS alignment is complete.

PSEU light ..... Extinguished

3.2. PREFLIGHT PROCEDURE

Aircraft status ..... Confirm

Light test ..... Test

Master LIGHTS test and dim switch – TEST

Use scan flow to check all lights flashing or illuminated. Use individual test switches or push to test feature to check appropriate lights which do not illuminate during the light test. The fire warning lights are not checked during this test.

Master LIGHTS test and dim switch – As desired

Oxygen ..... Test and set

Crew oxygen pressure – Check

Verify that the pressure is sufficient for dispatch.

Oxygen mask – Stowed

TEST/RESET switch – Push and hold

Verify that the yellow cross shows momentarily in the flow indicator.

Release the TEST/RESET switch. Verify that the yellow cross does not show in the flow indicator.

EFIS control panel ..... Set

MINIMUMS reference selector – BAROMETRIC

Select BAROMETRIC. Set MFRA to airfield elevation plus 1000 feet unless a different single engine flap retraction altitude is specified.

FLIGHT PATH VECTOR switch – As desired

METERS switch – As desired

BAROMETRIC reference selector – Set

Select barometric altitude reference. Set local altimeter setting.

VOR/ADF switches – Set

Mode selector – MAP

CENTER switch – As desired

Range selector – As desired

TRAFFIC switch – As desired

MAP switches – As desired

Ensure that Terrain and Airport are selected.

Clock ..... Set

TIME/DATE pushbutton - UTC time

NOSE WHEEL STEERING switch ..... NORM

Display select panel ..... Set

MAIN PANEL DISPLAY UNITS selector – NORM

LOWER DISPLAY UNIT selector – NORM

TAKEOFF CONFIG light
(if installed and operative) ..... Verify extinguished

CABIN ALTITUDE light
(if installed and operative) ..... Verify extinguished

Flight instruments ..... Set

Note: IRS alignment must be complete.

PFD – Correct

Flight mode annunciators – Blank

AFDS status is FLT DIR

Flight instrument indications are correct.

The NO V SPD flag is displayed until V–speeds are selected.

Verify no other flags displayed.

Altimeter – Set

ND – correct

Verify no flags displayed

Route – Displayed, correct

GROUND PROXIMITY panel ..... Check

FLAP INHIBIT switch – Guard closed

GEAR INHIBIT switch – Guard closed

TERRAIN INHIBIT switch – Guard closed

Verify that the GROUND PROXIMITY INOP light is extinguished.

GROUND PROXIMITY warning SYSTEM TEST switch..... Push momentarily

Perform full test on first flight of the day.

Verify switch guards down.

Verify proper operation of the following:

- BELOW G/S and GPWS INOP lights illuminate
- PULL UP and WINDSHEAR alerts illuminate
- “GLIDE SLOPE”, “PULL UP”, and “WINDSHEAR” aurals sound
- TERR FAIL and TERR TEST show on navigation displays
- terrain display test pattern shows on navigation displays
- terrain caution aurals sound and TERRAIN caution message shows on navigation displays.

Light controls ..... As desired

Audio control panel ..... Set

Seat ..... Adjust

FMC/CDU..... Set present position

Prior to dispatch, the crew must ensure that the Navigation Database is current and verify aircraft position is entered correctly. Use GPS L for position inputs. If this is unavailable, use GPS R position. A departure procedure shall not be used if doubt exists as to the validity of the procedure in the navigation database.

POS INIT page – Select

Use the GPS position from POS INIT page 2, enter present position on the SET IRS POS line. Confirm that the box prompts are replaced by the entered present position.

Note: Ensure selected GPS position agrees with current aircraft position.

IDENT page – Check

Verify airplane and engine MODEL and NAV DATA ACTIVE dates are correctly displayed.

RTE page – Select

Enter the departure and arrival four digit ICAO designator and the Virtual Ryanair callsign (RYRXXX) on RTE page 1.

Enter the ICAO routing from the OFP into the FMC RTE page.

Select and insert the expected SID and STAR, based on current and forecast weather. A departure procedure shall not be used if doubt exists as to the validity of the procedure in the navigation database.

Step through the route using the LEGS page and the EFIS Control panel PLN mode.

Enter relevant performance information.

When all this is complete, select “ACTIVATE”.

DEPARTURES page – Select

Select the active runway and SID.

PERF INIT page – Select

Verify total fuel quantity is displayed on the CDU and that the fuel quantity indicators agree, and are adequate for the planned flight. Enter zero fuel weight, fuel reserve (final reserve plus alternate fuel) and a cost index of 6. Enter EZFW if actual ZFW is not available to enable destination fuel prediction. Enter OFP fuel in the plan fuel data field. Enter optimum cruise altitude (or ATC capped level, if lower) and verify transition altitude. Enter top-of-climb wind (actual wind on OFP) and ISA deviation or top–of–climb OAT.

ARRIVALS page – Select

Select anticipated STAR and arrival instrument approach, if known.

Speed restrictions and route bypasses may be addressed at this point. 250 knots below FL100 applies to all flights unless removed by ATC when operating in Class A, B or C airspace only. Crews should consider turbulence and/or icing conditions when selecting a climb profile which achieves the most comfort and least exposure to icing conditions.

N1 LIMIT page – Select

Enter OAT

TAKEOFF page – Select

Insert Takeoff flap setting. Flap 5 is the standard takeoff flap setting unless performance considerations dictate otherwise.

FMC/CDU entries are complete to a point where loadsheet information is needed.

Flight control panel ..... Check

Verify that the standby hydraulic LOW QUANTITY light is extinguished.

Verify that the standby hydraulic LOW PRESSURE light is extinguished.

Verify that the STBY RUD ON light is extinguished.

ALTERNATE FLAPS master switch – Guard closed

ALTERNATE FLAPS position switch – OFF

Verify that the FEEL DIFF PRESS light is extinguished.

Verify that the SPEED TRIM FAIL light is extinguished.

Verify that the MACH TRIM FAIL light is extinguished.

Verify that the AUTO SLAT FAIL light is extinguished.

NAVIGATION panel ..... Set

VHF NAV transfer switch - NORMAL

IRS transfer switch – NORMAL

FMC source select switch (as installed) – NORMAL

DISPLAYS panel ..... Set

SOURCE selector - AUTO

CONTROL PANEL select switch – NORMAL

Fuel system ..... ___ KGS & 4/6 pumps ON

CROSSFEED selector – Verify operation

ENGINE VALVE CLOSED lights – Illuminated dim

SPAR VALVE CLOSED lights – Illuminated dim

FILTER BYPASS lights – Extinguished

CROSSFEED selector – Closed

VALVE OPEN light – Extinguished

Fuel quantity – Check

Verify total fuel quantity meets dispatch requirements. BLOCK Fuel + xxx kgs.

Cross check the fuel in the tanks against the BLOCK FUEL on the OFP

FUEL PUMPS switches (for tanks containing fuel) – ON

Center tank fuel pump switches should be positioned ON only if the fuel quantity in the center tank exceeds 453 kgs. The flight deck must be manned whenever center tank fuel pump switches are on.

LOW PRESSURE lights – Extinguished

CAUTION: **If a LOW PRESSURE light does not extinguish when the switch is positioned ON, position the switch OFF.**

Fuel System – Guidance

When checking the fuel system you should ensure that:

- Required pumps are on
- Crossfeed is serviceable and closed
- Fuel quantity indicators agree
- Total fuel quantity is displayed on the CDU
- Fuel contents are adequate for the planned flight (reference OFP.)

For aircraft with the new centre pumps installed (no placard), the center pumps may be turned on if the center tank contains more than 453 kgs. If the captain deems that the amount of fuel in the centre tank is not sufficient to keep the fuel low pressure lights out during T/O then he may elect to leave the centre tank pump switches off until passing FL 100

For aircraft with the old centre pumps installed (placarded) the centre pumps should not be switched on for T/O with less than 2,300kgs of fuel in the centre tank.

CAB/UTIL & IFE switches (as installed) ..... ON

Note: Both switches must be ON on vRYR airplanes.

GALLEY power switch (as installed) ..... ON

Electrical system ..... Set

STANDBY POWER switch – AUTO (guard down)

Generator drive DISCONNECT switches – Guards down

BUS TRANSFER switch – AUTO (guard down)

CIRCUIT BREAKER and PANEL light controls ..... As desired

EQUIPMENT COOLING switches ..... NORMAL

OFF lights – Extinguished

Passenger signs ..... Set

FASTEN BELTS switch – ON

Windshield WIPER selectors ..... PARK

WINDOW HEAT switches ..... ON

Position switches ON at least 10 minutes before takeoff.

OVERHEAT lights – Extinguished

ON lights – Illuminated (except at high ambient temperatures)

PROBE HEAT switches ..... OFF

Verify that all lights are illuminated.

WING and ENGINE ANTI – ICE switches ..... OFF

VALVE OPEN lights – Extinguished

Hydraulic pump switches ..... ON

System A HYDRAULIC PUMPS switches – ON

System B HYDRAULIC PUMPS switches – ON

Electric pump LOW PRESSURE lights – Extinguished

Brake pressure – 2800 PSI minimum

MFD SYSTEM switch – Push

System A and B pressure – 2800 PSI minimum

Quantity indicators – No RF indication displayed

Pressurization indicators ..... Check

Cabin differential pressure – Zero

Cabin altitude – Field elevation

Cabin rate of climb – Zero

Air conditioning system ..... Both packs, bleeds ON

AIR TEMPERATURE source selector – As desired

TRIM AIR switch – ON

Temperature selectors – As desired

RAM DOOR FULL OPEN lights – Illuminated

RECIRCULATION FAN switches – AUTO

Both Recirc fans shall be in Auto and left in Auto throughout the day. There is no requirement for the crew to switch these to OFF unless following a QRH procedure.

Air conditioning PACK switches – Both AUTO or HIGH
ISOLATION VALVE switch – OPEN
Engine BLEED air switches – ON
APU BLEED air switch – As required
ON unless external air is used for start.

Pressurization system ..... Set

FLIGHT ALTITUDE indicator – Filed level or lower if performance limited or level capped.

LANDING ALTITUDE indicator – Destination field elevation

Pressurization mode selector ..... AUTO

AUTOMATIC FAIL light – Extinguished

Exterior light switches ..... Set

Navigation lights must be on at all times when AC power is connected. Wheel well lights will be selected on for walkarounds.

Logo lights will be on during the hours of darkness below FL100.

Caution when using wing lights as they can cause distraction to other ground traffic.

Ignition switches ..... IGN L or R

Select IGN R when operating through manned maintenance stations.

ENGINE START switches ..... OFF

Mode control panel ..... Set

When selecting a value on the MCP, ensure the corresponding display on the instrument panel changes, if applicable.

COURSE(S) – Set and crosscheck

FLIGHT DIRECTOR switches – ON

Altitude – SID clearance altitude. If clearance has not been received, set expected SID clearance altitude plus 100 feet.

Autopilots – Disengaged

Integrated standby flight display ..... Check

Approach mode display – Blank

Set local altimeter setting

Verify flight instrument indications are correct

Verify no flags or messages are displayed.

Engine display control panel ..... Set

N1 SET selector – AUTO

Permits FMC control of N1 bugs.

FUEL FLOW switch – RATE

Move switch to RESET, then RATE.

SPEED REFERENCE selector – AUTO

Permits FMC control of reference speed bugs.

AUTO BRAKE select switch ..... RTO

AUTO BRAKE DISARM light – Extinguished

ANTISKID INOP light ..... Extinguished

Engine instruments ..... Check

MFD ENGINE switch – Push

Note: EGT, F/F, oil pressure and oil temperature pointers and digital readouts are not displayed until the start switch is moved to GRD.

Primary and secondary engine indications – Normal

- engine indications display existing conditions
- no exceedance values are displayed
- engine oil quantity meets dispatch requirements (12 quarts.)

SPEED BRAKE lever ..... DOWN detent

Advance thrust levers to forward stop to ensure takeoff configuration warning horn sounds. Retard thrust levers to idle stop. Ensure the speed brake lever is in the DN detent.

Reverse thrust levers ..... Down

Forward thrust levers ..... Closed

Start levers ..... CUTOFF

Parking brake ..... Set

Parking brake warning light – Illuminated

STABILIZER TRIM cutout switches ..... NORMAL

Wheel well fire warning system ..... Test

Verify fire warning bell sounds, master FIRE WARNING lights, MASTER CAUTION lights and OVERHEAT/DET annunciator illuminate.

Fire warning BELL CUTOUT switch – Push

Verify WARN lights and fire warning bell cancel.

WHEEL WELL fire warning light – Illuminated

Audio control panels ..... Set

PANEL OFF lights – Extinguished

Set panels – As desired

VHF NAVIGATION radios ..... Set for departure

[Set VHF navigation radios to ensure in the event of FMC failure that conventional navigation may be resumed.]

ADF radio ..... Set

FLOOD and PANEL light controls ..... As desired

Weather radar ..... Set

Transponder ..... Set

RUDDER and AILERON trim ..... Free & zero

Check trim for freedom of movement, set trim at zero units.

STABILIZER TRIM override switch ..... NORMAL

Complete the Route Check, Instrument Crosscheck, Takeoff Briefing, Initial Emergency Briefing (when appropriate), Taxi and SID briefing.

Call “BEFORE START CHECKLIST DOWN TO THE LINE.”

3.3. FINAL CDU PRE-FLIGHT PROCEDURE

When the loadsheet arrives at the airplane:

FMC/CDU ..... Set

PERF INIT page – Select

Enter zero fuel weight and confirm FMC gross weight agrees with the loadsheet.

EXECUTE.

N1 LIMIT page – Select

Confirm the maximum rated N1 setting.

Enter the selected derate and confirm the N1 setting.

Enter and confirm the assumed temperature which reflects the maximum selected reduction available for takeoff. Assumed temperature thrust reduction shall be used with the exception of the following:

- contaminated or slippery runway
- windshear is suspected
- anti-skid inop
- thrust reverser inop
- TOW is within 500 kgs of the RTOW (1000kgs when obstacle limited)
- EEC in alternate mode
- landing gear extended flight
- bleeds off takeoff
- when prohibited by airfield brief
- crosswind exceeding 10 knots.

Note: When crosswind in excess of ten knots exists and no assumed temperature reduction off a given derate is possible, then the next highest derate shall be selected, i.e., 22k derate, no assumed temp possible, select 24k for this takeoff. This is a tailstrike prevention policy.

Thrust mode display – Check

Verify dashes are displayed.

TAKEOFF REF page 1/2 – Select

Enter takeoff flap setting.

Enter V1, VR and V2.

TAKEOFF REF page 2/2 – Select

Enter thrust reduction altitude if different from displayed value.

Takeoff data ..... Review

Verify preflight complete and scratchpad is clear except for < INDEX.

Stab trim ..... Set

Confirm trim setting required after appropriate corrections for thrust setting used and set this stab trim setting.

IAS bugs ..... Set

Verify V1 speed is displayed at the top of airspeed indication. Set V2 in the MCP IAS/Mach display.

Prior to pushback the select the TAKEOFF page on Left CDU and the LEGS page on the Right CDU.

3.3.1. FINAL CDU PRE-FLIGHT PROCEDURE – GUIDANCE

Once the Loadsheet arrives, the captain will check its accuracy, sign it and give a copy to the handling agent as soon as possible in order to allow the handling agent to vacate the aircraft and continue with their own departure preparations.

The captain will call out the ZFW and the F/O will input the data into the FMC PERF INIT, crosschecking the TOW.

The F/O will have the RTOW book open and will confirm the correct pages, as per T/O briefing. Both pilots will check for accuracy (airport, runway, ET, flap, engine rating and wet/dry.)

The Assumed Temperature Thrust Reduction, if applicable, is the maximum selected reduction available for takeoff.

RTOW speeds must be used.

It is acceptable to line select “wet speed” ONLY on TAKEOFF page 2, not SKR-R. FMC speeds may be used if they differ from the calculated RTOW speed by no more than 1 knot. The co-pilot will record the performance calculations on the Flight Envelope.

Once speeds have been agreed and entered, the F/O will select the V2 on the MCP.

The captain states the calculated stab trim and records the value on the Loadsheet. The F/O sets this value.

The PF selects the TAKEOFF page and the PM selects the LEGS page for departure.

A simple Gross Error Check is to simply take the aircraft weight as 42,000kgs and each passenger as 100kgs. The sum of these will approximate the ZFW on the Loadsheet.

A more detailed method of checking the accuracy of the figures is subtract the actual load from the allowed load. Now add the resulting figure to the T/O weight. This should give you the max allowable T/O weight.

3.4. INITIAL EMERGENCY BRIEFING

"Above 80kts takeoff will only be rejected for:

- Fire or fire warning
- Engine failure
- Predictive windshear warning
- Airplane is unsafe or unable to fly."

"Before V1, on rejected takeoff I will:

- Simultaneously close the thrust levers and disengage the autothrottle
- Apply maximum manual braking or verify operation of RTO autobrakes
- Manually raise the speedbrake lever
- Apply maximum reverse thrust and stop the aircraft. I will set the parking brake.
- Select flaps 40 when the parking brake has been set
- Inform ATC of the reject."

Very few RTOs result in passenger evacuation. Flight Safety Foundation figures suggest that approximately 90% of evacuations are performed without due cause and that almost 100% of evacuations involve death or serious injury. It follows that an evacuation is not a 'default option' but is in fact a dangerous maneuver which should be performed only when the situation truly demands it. Good airmanship requires that all opportunities are taken to obtain and evaluate any available information prior to initiating an evacuation such that the best possible decision is reached. It further follows that, since evacuations will only thus be initiated for sound reasons, they should always be followed through to the completion of all QRH actions.

3.5. FINAL FLIGHT DECK PREPARATION

There have been occurrences where a crew has received Start Clearance only, with no clearance to Push. Crews have responded to the clearance to Start with the full Push and Start procedure and pushed from stand without permission. Avoid responding to clearances with rhyming responses and ensure that the actual clearance received is complied with.

A flashing anti-collision light will prevent the movement of road traffic in the vicinity of the aircraft. Sitting on the ramp with a beacon rotating may impede other airport staff from getting on with their jobs so the selection of anti-collision light ON should not be selected unless ATC clearance has been received.

Flight deck access system switch ..... NORM

Flight deck windows/cabin door ..... Locked

Transponder ..... ALT OFF

Engine start clearance ..... Obtain

Hydraulic A pumps ..... As required

Pushback shall always be carried out with the hydraulic A pumps off. If the nose gear steering lockout pin is not installed or is unavailable, the engine start will be delayed until the towbar has been disconnected.

CAUTION: Do not hold or turn the nose wheel steering wheel during pushback or towing. This can damage the nose gear or towbar.

Doors ..... Closed

All exterior door annunciator lights – Extinguished

Air conditioning PACKS ..... OFF

ANTI COLLISION light ..... ON

Parking brake ..... Set

Call and complete “BEFORE START CHECKLIST.”

3.6. BEFORE TAXI PROCEDURE

The Before Taxi Procedure is fundamental to the safe operation of the flight. Crucial systems and, most of all, the aircraft configuration, are set up during this procedure. Always use the laminated Normal Checklist and follow the checklist physically until it is complete. Do not complete these from memory. This is a crucial phase in our prevention of a Take-Off Configuration error during the upcoming take-off.

GENERATOR 1 and 2 switches .....ON

PROBE HEAT switches ..... ON

WING ANTI–ICE switch ..... As needed

ENGINE ANTI–ICE switches ..... As needed

Hydraulic A pumps ..... ON

PACK switches ..... AUTO

ISOLATION VALVE switch ..... AUTO

APU BLEED air switch ..... OFF

ENGINE START switches ..... CONT

APU switch ..... OFF

Recall ..... Check

Verify that all system annunciator panel lights illuminate and then extinguish.

Verify that the ground equipment is clear.

Flight controls ..... Check

Make slow and deliberate inputs, one direction at a time.

Move the control wheel and the control column to full travel in both directions and verify:

- freedom of movement
- that the controls return to center

Hold the nose wheel steering wheel during the rudder check to prevent nose wheel movement.

Move the rudder pedals to full travel in both directions and verify:

- freedom of movement
- that the rudder pedals return to center

Flap lever ..... Set takeoff flaps

Flap position indicator and FLAP lever – Set for takeoff

LEADING EDGE FLAPS EXTENDED green light – Illuminated.

When accomplishing the before taxi checklist the Captain shall check the FMC TAKEOFF page and verify the required flap setting. The Captain will physically check that the flap lever is in the required position and that the flap position is correctly indicated and that the green LE FLAPS EXT light is illuminated.

Stabilizer trim ..... ____UNITS REQ'D, ____SET

Verify stabilizer trim is set for takeoff.

When accomplishing the before taxi checklist the captain must verify that the stab trim is set for takeoff.

Engine start levers ..... IDLE detent

Blank the lower display unit.

Update changes to the taxi briefing, as needed.

Call and complete "BEFORE TAXI CHECKLIST."

Obtain taxi clearance and confirm that ground obstacles are clear on the right-hand by calling "CLEAR RIGHT", confirm the left-hand side is clear by calling "CLEAR LEFT", select taxi and turnoff lights to on.

It is imperative that ATC taxi instructions are understood by pilot before taxi is commenced.

4. TAXI AND TAKEOFF

4.1. TAXI PROCEDURES

Taxiing the aircraft is a crucial phase of flight when many critical issues are dealt with and resolved.

Due to the restrictive nature of some of Virtual Ryanair's destination aprons, keep breakaway thrust to a minimum. Normally, 30 to 35% N1 is all that is necessary to commence taxi.

Once the taxi clearance has been received, release the parking brake and momentarily advance the thrust levers forward until the white N1 Command Sector is at 70% and immediately reduce to the desired N1 for commencement of taxiing.

During this config check, actual N1 will only begin to increase and will not reach more than 20-25% N1 when the action is carried out promptly. Except at very light weights the aircraft will not move, however any existing takeoff configuration triggers, if present, will set off the aural warning.

If a warning occurs, the captain shall reduce thrust to idle, stop the aircraft if it is in motion and set the parking brake. The cause of the TOC warning must be positively identified and all possible causal factors checked and appropriate checklists repeated in full before taxi is recommenced. The second attempt to taxi will be initiated as described above.

Maximum ground speed in a straight line is 30 knots, 15 knots on the apron and 10 knots around corners. Maximum ground speed whilst taxiing on slippery surfaces is 10 knots and 5 knots around corners. On contaminated taxiways and runways max taxi speed is 5 knots. These speeds are maximum and captains must exercise sound judgment when other factors may require slower speeds.

Do not cut corners. If the taxi line is displaced, follow this taxi line with the nose wheel. If the taxi line is not displaced, allow the nose gear to overshoot turns to ensure the main gear straddles the taxiway centerline. The nose gear may be displaced from the centerline to avoid airport lighting. This will also reduce wear and tear and enhance passenger comfort. Be aware that the geometry of the winglet scribes the greatest arc during turns.

Never use thrust against the brakes. Use brakes to slow the aircraft down to the desired taxi speed and then release.

In low visibility the Captain will normally follow the green lights on the taxi routing. If this lighting is not available, confirm taxiways with airport signage and HDG on flight instruments. If unfamiliar with the airport or route, consider requesting progressive taxi or a Follow Me vehicle and stop the aircraft if in doubt.

Always use standard ATC radio phraseology and read back all clearances. Avoid distractions during critical taxi phases.

4.2. BEFORE TAKEOFF PROCEDURE

The most important message to convey to a crew during this phase of our operation is: DO NOT RUSH. A few moments to ensure that this procedure has been completed properly may save hours of paperwork, embarrassment and possibly worse. It is essential that the taxi progress is monitored and any errors, especially incorrect routings and runway incursion possibilities are noted and rectified.

At all times, changes to the active clearance must be addressed cautiously and with the confirmation of the captain. If there is a change to the planned SID, crews must stop the aircraft and NOT recommence taxi or accept line up or takeoff clearance until rebriefing and re-confirming all SID FMC entries. This includes performance, NAVAID and MCP selections for the departure.

Be aware that the new route may be significantly longer than the original and additional fuel uplift may be necessary.

Commencing taxi the captain will call for the Before Takeoff checklist when the aircraft is clear of obstructions. This command is the cue to begin the Before Takeoff checklist. Reading of the checklist may be delayed until clear of the apron.

Config ..... checked

The Captain will call “Config” and advance the thrust levers forward until the white N1 Command Sector is at 70% and immediately reduce to the desired N1. Verify takeoff configuration warning horn does not sound.

Flaps ..... ___ Green light

Fuel quantity (center tank) ..... Check

Both center fuel tank pump switches may be OFF for takeoff if center tank fuel is less than 500 kilograms to avoid nuisance fuel alerts during the takeoff phase. If placarded, with less than 2300kg in the center tank, CTR Tank Pumps OFF for takeoff.

Stabilizer trim ..... ____UNITS SET

Verify stabilizer trim is set for takeoff.

Takeoff briefing ..... Captain review

Items to review include, but are not limited to:

- Packs
- Bleeds
- Speeds
- SID - include initial turn requirement
- Stop altitude - The captain shall first refer to the FMC LEGS page, then point to and call the hard altitude previously noted or inserted. The captain shall then point to the selected MCP altitude and confirm that it is the same value as on the FMC LEGS page and call “SET.”

Cabin.....Secure

Before entering the departure runway, verify that the runway and runway entry point are correct.

Prior to entering the runway, the crew will confirm that the runway and approach are clear. Seat the cabin crew by announcing on the PA: "CABIN CREW, SEATS FOR DEPARTURE," select strobes ON, A/T to ARM, ARM LNAV (as required) and transponder to TA/RA.

MCP: The Capt will select Radar and the F/O will select TERR. Confirm that the intended departure routing is clear of adverse weather and that the intended departure routing is clear of terrain

The captain will turn off the taxi light.

MCP ..... Set

Transponder ..... TA/RA

Fixed landing and strobes ..... ON

Only when takeoff clearance has been received.

Retractable lights ..... ON

Complete the BEFORE TAKEOFF checklist.

4.3. TAKEOFF PROCEDURE

It is preferable to conduct a rolling takeoff as this greatly reduces the possibility of FOD without significantly affecting takeoff performance.

The standard Virtual Ryanair procedure is that the captain should have the weather radar on and the F/O Terrain selected for T/O. However, this can vary if the captain deems it necessary to have both on weather radar or Terrain for departure.

MAP mode shall be used on both sides for departure.

Advance the thrust levers to approximately 40% N1. Engine RPM increase may vary considerably between the engines, particularly when there is a crosswind. The difference in acceleration rates between engines is usually much more noticeable at RPMs below 40% N1. The time taken for spool up from IDLE to 40% can differ by several seconds between the two engines. If TOGA is pushed before engine RPMs are at 40% on BOTH engines there may be a significant thrust asymmetry and there could be difficulties with directional control. Ensure both engines N1 RPMs are at 40% before calling “STABILISED”. Push TOGA and call “SET TAKEOFF THRUST.”

Once the target N1 has been reached, compare actual N1 with FMC TAKEOFF pages, and call “TAKEOFF THRUST SET.”

Check the engine instruments on the upper DU and check for any pop-ups on the lower DU and once satisfied call “INDICATIONS NORMAL.”

When 80 knots is indicated on the speed-tape the PM will call “80 KNOTS”. Call “V1” and ensure that the V1 call is completed as V1 is reached. Call “ROTATE” at the agreed Vr. PF must commence rotation at the agreed Vr regardless of whether the PM has called it or not.

> It is preferable to conduct a rolling takeoff as this greatly reduces the possibility of FOD without significantly affecting takeoff performance. Further, engine surge can occur if T/O thrust is set prior to brake release. Therefore it is strongly advised that a rolling T/O procedure is used even in crosswind conditions.
>
> With very strong headwinds, the thrust levers may not advance to T/O N1 as Throttle Hold will have engaged too early. In this case, manually advance the thrust levers as required.

4.3.1. TAKEOFF ROLL AND ROTATION

Rotate at a rate of 2 to 3 degrees per second. Beware of a “dead band” around 10 degrees, and maintain a constant rate of rotation towards 15 degrees nose up to a pitch limit of 20 degrees.

Flaps 5 are the standard flap setting for all departures. If there is a crosswind of 10 knots or more, then no assumed temperature reduction is allowed.

Flaps 1 departures are only authorized for performance purposes.

Once airborne, accurately follow the F/D command bars matching the HDG if required.

4.3.1.1. TAKEOFF ROLL AND ROTATION – GUIDANCE

A slight forward pressure on the control column should be applied until reaching 80 knots. Above 80 knots, relax the forward pressure towards neutral. For optimum T/O and initial climb performance, initiate a smooth continuous rotation at Vr towards 15 degrees pitch up.

Tailstrike prevention procedures are discussed below and it must be stressed that a proper rotation at the correct rate will not only avoid tailstrike, but also gives the best aircraft performance.

4.3.1.2. CROSSWIND TAKEOFFS – GUIDANCE

As the aircraft accelerates the crosswind will begin to push the tail which results in the nose veering upwind and the aircraft tracking left or right of the centre line. Use rudder to counteract this tendency. It is important to use smooth rudder pedal deflection as pumping the pedal will make directional control difficult.

Maintain wings approximately level during the takeoff roll by applying control wheel displacement into the wind. During rotation, continue to apply control wheel in the displace position to keep the wings level during lift-off. The airplane is in a sideslip with crossed controls at this point. A slow, smooth recovery from this sideslip is accomplished after lift-off by slowly neutralizing the control wheel and rudder pedals.

For Take-off, 1 unit (6 degrees of aileron deflection) of into wind aileron is normally sufficient. More than 1.5 units of control wheel deflection (10 degrees of aileron deflection) will cause the Spoilers to deflect which in turn affects take- off performance. Spoiler deployment at this stage also increases the potential for tail strike.

Do not trim during rotation.

4.3.1.3. TAILSTRIKE PREVENTION – GUIDANCE

Be aware of the following threats that increase the likelihood of tailstrike during takeoff:

- Improperly trimmed stabilizer
- Improper rotation technique - in gusty conditions you rotate at a speed equal to if not greater than VR. Use the normal rate of rotation.
- Avoid rotation during a gust. If a gust is experienced near VR, as indicated by stagnant airspeed or rapid airspeed acceleration, momentarily delay rotation.
- Incorrect Takeoff speeds - this has the same effect as early rotation.
- Excessive initial pitch attitude.
- Excessive use of aileron - this will cause the spoilers to rise which increases drag, reduces lift and leads to over-rotation.
- Tailstrikes increase significantly with crosswinds in excess of 20 knots.

Tailstrikes that have occurred in Virtual Ryanair were in gusty wind and strong crosswind conditions.

Many pilots mistakenly believe that during these conditions the greatest threat to a well-controlled and safe takeoff is directional control. Accordingly it is not uncommon to see pilots put in large control wheel displacements (and excessive forward pressure) during the takeoff roll. This is not the correct technique and, when applied, greatly increases the possibility of a Tailstrike.

The FCTM is very clear in this area. Any reading of this document will confirm that the greatest threat to a well conducted and safe take-off in strong and gusty crosswind conditions is a Tailstrike.

Gusty Wind and Strong Crosswind Conditions

ALL of this section of the FCTM is crucial to understanding how to avoid a Tailstrike in these conditions but the following is particularly relevant to this memo: -

To increase tail clearance during strong crosswind conditions, consider using a higher VR if takeoff performance permits..... Avoid rotation during a gust. If a gust is experienced near VR, as indicated by stagnant airspeed or rapid airspeed acceleration, momentarily delay rotation.

".....Do not rotate early or use a higher than normal rotation rate in an attempt to clear the ground and reduce the gust effect because this reduces tail clearance margins. Limit control wheel input to that required to keep the wings level. Use of excessive control wheel increases spoiler deployment which has the effect of reducing tail clearance. All of these factors provide maximum energy to accelerate through gusts while maintaining tail clearance margins at liftoff."

In Summary

- If extra control wheel displacement is required to maintain the wings "approximately" level in gusty wind and strong crosswind conditions, limit the extra amount to the minimum required to achieve this.
- In gusty wind and strong crosswind conditions delay the rotation until VR for the performance limited gross weight rotation speed. Takeoff performance is assured using this method.
- Do not rotate early or use a higher than normal rotation rate because this reduces tail clearance margin.

5. CLIMB

5.1. AFTER TAKEOFF PROCEDURE

The After-take off phase contains one of the last chances to catch an improperly configured Air Conditioning and Pressurization System. This system could also be called the Passenger Comfort and Life Support System so important is it in ensuring a pleasant experience for our passengers and to ensure their safety. Again, DO NOT RUSH this check. Below is sound guidance as to how to complete the check. Use the guidance meticulously to ensure a safe flight. Lookout is important during this phase, especially at those airports marked 'VFR traffic' on the RTOWs. A good lookout is more important than expediting the after-take-off procedure.

During flap retraction, the speed must not only be at the maneuvering speed for the existing flap setting, but also showing an accelerating speed trend vector.

This is important in order to ensure that the aircraft is accelerating to the next flap speed as the flaps are retracting.

Do not engage V/S until the flaps are up.

After "Flaps Up No Lights" an MCP climb selection should be made and the After Takeoff checklist called for. The Standard climb mode is VNAV.

If using VNAV, then any departure speed or altitude restrictions should be considered. The pilot who sets the new MCP ALT should select ALT INTV if selected altitude exceeds the altitude restrictions in the FMC CLB page and if an unrestricted climb has been approved by ATC.

SPD INTV may be used to control speed when in VNAV at this stage of the flight. During the after T/O Procedure the Air Cond & Press check should be read aloud

The Altimeters check is a challenge and response check.

It is Virtual Ryanair's preferred policy to engage the A/P when the wings are level. Engaging the A/P in a turn frequently results in a "Bank Angle" EGPWS warnings and OFDM alerts. During complex SIDs it may be preferable to engage the autopilot at 1000/1500 feet while being aware of the possibility of an Overbank.

In busy TMA's, uncontrolled airspace or with weather/terrain/local traffic considerations, the autopilot must be used in order to improve the situational awareness of the PF.

Naturally you should respond to any ATC calls in good time.

5.1.1. AFTER TAKEOFF PROCEDURE - NADP 1

Pilot Flying	Pilot Monitoring
Maintain a minimum of V2 + 15 knots during initial climb. At light gross weight a higher speed (up to V2 + 25) may be selected, to synchronize F/D pitch command and avoid objectionable body attitude. Do not exceed 20 degrees nose up.	Monitor engine instruments and cross–check flight progress.
Above 400 feet, call for appropriate roll mode, if required. Verify proper mode annunciation.	Select/verify roll mode. Verify proper mode annunciation.
At 1,500 feet AAL, verify the FMA displays N1 and select CMD A or B. Call “COMMAND A or B.”	At 1500’ AAL, verify automatic thrust reduction to climb thrust.
At 3000 feet AAL call “BUG UP” and set the airspeed bug to “UP.”	
Retract flaps on schedule.	Position FLAP lever as directed and monitor flaps and slats retraction. Call “FLAPS UP, NO LIGHTS.”
Above 3,000 feet AAL, engage VNAV or select normal climb speed and verify annunciation.	Verify proper mode annunciation.
Call “AFTER TAKEOFF CHECKLIST” when flaps are up.	Position landing gear lever OFF, engine start switches as required. Verify air conditioning and pressurization are operating and configured normally with a positive pressure differential. Complete the AFTER TAKEOFF checklist.

5.1.2. AFTER TAKEOFF PROCEDURE - NADP 2

Pilot Flying	Pilot Monitoring
Maintain a minimum of V2 + 15 knots during initial climb. At light gross weight a higher speed (up to V2 + 25) may be selected, to synchronize F/D pitch command and avoid objectionable body attitude. Do not exceed 20 degrees nose up.	Monitor engine instruments and cross–check flight progress.
Above 400 feet, call for appropriate roll mode, if required. Verify proper mode annunciation.	Select/verify roll mode. Verify proper mode annunciation.
At 1,000 feet AAL, select CMD A or B. Call “COMMAND A or B.” call “BUG UP” and set the airspeed bug to “UP.”	
Retract flaps on schedule. Maintain “UP” speed until reaching 3000 feet AAL.	Position FLAP lever as directed and monitor flaps and slats retraction. After calling “SPEED CHECKS, FLAPS UP” call ATC as required.
At 1500 feet AAL, verify the FMA displays “N1.”	At 1500 feet AAL, verify automatic thrust reduction to climb thrust.
Call “AFTER TAKEOFF CHECKLIST” when flaps are up and above 3000 feet AAL.	Above 3000 feet AAL, call “FLAPS UP, NO LIGHTS.” Position landing gear lever OFF, engine start switches as required. Verify air conditioning and pressurization are operating and configured normally with a positive pressure differential. Complete the AFTER TAKEOFF checklist.
Above 3,000 feet AAL, engage VNAV or select normal climb speed and verify annunciation.	

5.2. TAKEOFF FLAP RETRACTION SPEED SCHEDULE

T/O FLAPS	SELECT FLAPS	AT: (For all weights)
25	15 5 1 UP	V2+15 "15" "5" "1"
15	5 1 UP	V2+15 "5" "1"
10	5 1 UP	V2+15 "5" "1"
5	1 UP	V2+15 "1"
1	UP	"1"

- "UP" – Flaps up maneuvering speed.
- "1", "5", "10", "15". "25" – Number

The response "Speed checks, flaps__", requires the PM to ensure that the current speed is at or above the maneuvering speed for the existing flap setting and showing an accelerating speed trend vector.

Note: Limit bank angle to 15 degrees until reaching V2 + 15.

Note: When using NADP 1 and ATC or the assigned SID requires a level off altitude of less than 3000 feet, the PF will call "BUG UP" at ALT ACQ and retract the flaps on schedule.

Note: The A/P may be engaged at a later stage of flight if desired. The earliest opportunity is at 1000 feet (NADP 2) or 1500 feet after the A/T engages in N1 (NADP 1). If a later engagement is planned, the standard callout at 1000/3000 feet AGL by the PF is "BUG UP." The PM will position the fly speed bug to UP and complete flap retraction on schedule at the PF's command.

Note: V/S should not be used before the flaps are up.

5.3. ALTIMETRY SETTING PROCEDURES

SOP discipline is essential when changing altimeter pressure settings.

Flap Retraction and Altimeters checks are done separately. Both are crucial and complex procedures that do not need to be completed simultaneously. The SOP states that "both pilots may set their altimeters to Standard" above three thousand feet. This procedure has been put in place to cater for departures where there are low Transition Altitudes and low stop-heights in a SID.

Altimeters must be set to airfield QNH for takeoff.

When cleared to climb above transition altitude, and the aircraft is above 3000 feet AAL, both pilots may set their altimeters to Standard.

The standby altimeter is set to Standard when climbing through MSA. While the standby altimeter will normally be changed when above the MSA.

The maximum difference between the captain's and first officer's altimeter whilst flying in RVSM airspace is 200 feet. Should a differential exist, the difference will be split and ALT HLD mode will be used to control the airplane flight path.

On receiving clearance to descend to an altitude, both pilots will set QNH. Caution must be exercised should an intermediate level off above transition level be issued by ATC in which case STD must be re-selected.

5.4. CLIMB AND CRUISE PROCEDURE

PILOT FLYING	PILOT MONITORING
Set altimeters to STD when cleared above transition altitude and above 3000 feet AGL. If the PF has not set STD on passing transition altitude, the PM reverts to the standard callout "ALTIMETERS." Both crewmembers shall then set STD.	
Response: Call the exact passing level/altitude. "PASSING ___CLIMBING___, TEN CHECKS."	Challenge: Call "ALTIMETERS" when passing FL100/10,000 feet. Ten checks are performed aloud as follows: "LIGHTS, APU, FASTEN BELTS, AIRCOND AND PRESS, RECALL."
	Verify pressurization normal at FL200, 300 and 400 during the climb phase.
When cleared to final cruise level, confirm MCP, FMC and flight altitude selection are in agreement. Call "FL___SET THREE TIMES."	"CHECK"
Ensure that a maximum 1000 feet per minute rate of climb is not exceeded during the last 1000 feet when in RVSM airspace. It is generally preferred that VNAV is left engaged. Follow RVSM procedures for the remainder of the flight.	
Call for enroute NAVAID selections, as appropriate.	Select NAVAIDS.
	Position the appropriate center tank fuel pump switch OFF when the pump LOW PRESSURE light illuminates.
Set MCP altitude selector for descent. Verify FMC selections.	Prior to top of descent, select and verify the planned arrival procedure on the FMC.
At top of descent point observe descent initiated and verify proper mode annunciation.	

5.5. FL100 AND "10 CHECKS"

The Press and Air Conditioning check in this sequence is the last chance for picking up an improperly set Life Support System. Crews who have missed this opportunity and subsequently encountered depressurizations have regretted not doing this check properly. Take your time and make sure what is being displayed by the system indicators is fully understood. It is SOP and good airmanship to check these items not only at FL100 but also 200, 300 and 400 in both climb and descent.

There shall be no unnecessary conversation or operational paperwork from the parking stand to FL100 and from FL100 to the arrival parking stand.

The PM should complete a silent scan of the overhead panel paying particular attention to the Air Conditioning and Pressurization systems every 10,000 feet.

Once cleared to cruise level the PF will check that cruise altitude is set "3 times" i.e: -

- MCP Alt window
- FMC CLB page
- Press panel Flt Alt.

Once through FL100, the centre pumps should be switched on. When the fuel decreases to 950 kgs, for aircraft with centre pump restrictions, the crossfeed valve should be opened and the RH pump switched off. At the first indication of a centre tank fuel low pressure light, the pump should be switched off and the crossfeed valve closed.

Note: When established in a level attitude at cruise, if the center tank contains usable fuel and the center tank pump switches are off, the center tank pump switches may be positioned ON again. If the center tank contains more than 453 kgs, the center tank pump switches must be positioned ON. Verify the LOW PRESSURE lights extinguish. Position the appropriate center tank fuel pump switch OFF when the LOW PRESSURE light illuminates.

6. CRUISE

6.1. R/T DISCIPLINE

Correct ATC terminology is essential. Accurate read back of all clearances and instructions especially when using the terms “Heading,” “Altitude” and “Flight Level” will ensure that there is no confusion with ATC and therefore reduce the amount of time spent speaking on the frequency. Always use the correct full callsign.

Before leaving a frequency, it is important to pause in order to allow ATC the chance to correct your frequency read back. On selecting a new frequency, it is important to pause before speaking in order to not block ATC or other aircraft.

6.2. CLIMB AND CRUISE PROCEDURES

MCP altitude changes following an ATC instruction:

(1) ATC give instruction to change ALT/Level
(2) Respond to ATC
(3) Reset MCP altitude and call “SET”
(4) Verify the correct MCP altitude is displayed on the PFD and call “XXX CHECKED.”.

For unrestricted climb it is desirable for economy reasons to delete Climb 1 or Climb 2 in the FMC N1 page.

Anticipate the use of engine/wing anti-ice during all phases of flight. Monitor the TAI indications on the upper DU after the engine anti-ice switches have been selected ON.

Flight crews will reduce rate of climb to 2000 FPM at 2000 feet to level-off and 1000 FPM at 1000 feet to level-off. This restriction will be observed below FL200 when operating in busy TMA’s or high density traffic areas. This policy is in order to avoid unnecessary TCAS warnings. However, it must be stressed that this does not relieve the crew of the requirement to adhere to any ATC altitude or speed restrictions. Equally, it is acceptable to monitor VNAV performance if it is within the above criteria.

When within 1000 feet of the cleared altitude there should be no distractions in the flight deck and the ALT ACQ closely monitored.

Maximum use of the AFDS is recommended in busy TMA’s and uncontrolled or class D and below airspace in order to ensure a good lookout and situational awareness.

During cruise, monitor UIR/FIR boundaries for correct handover by ATC.

6.3. CRUISE PROCEDURES

No later than top of Climb, the PM shall enter the destination QNH from the METAR on the 'landing card' section of the loadsheet. This serves as a gross error check when the crew reaches the destination and are given the QNH from ATIS or ATC.

The METAR is always entered on the loadsheet, ATIS is always entered on the PFD QNH pre-selector.

If there is a difference between the METAR and the QNH given by ATIS, then the crew must question this and confirm the actual QNH with ATC.

If a non-precision approach is being flown the crew MUST confirm the QNH with ATC before commencing the approach.

RVSM Altimeter Performance Record:

> At intervals of approximately one hour, crosschecks between the primary and secondary altimeters shall be made. The actual altitudes shall be recorded on the OFP. Both altimeters need to agree within +/- 200 feet. Failure to meet this condition will require that the altimeter system be reported as defective and notify ATC.

Enroute Weather:

The "Plan to land at the nearest suitable airport" contingency is partly fulfilled by awareness of the nearest suitable airport. This may be obvious when there is widespread good weather in an area or shall be ascertained by obtaining enroute weather reports. There is no requirement to obtain enroute weather when the weather conditions are good.

In all cases, the preference will be to get weather for enroute Virtual Ryanair bases as these are the most likely enroute diversion airports. Use the ND to pick the airports that are required and then the Low Level enroute chart to find the ATIS. The frequency is listed beside the airport ident.

6.4. IMMEDIATE LEVEL OFF PROCEDURE

There have been situations in which the crew has been instructed without notice to level off at the present or rapidly approaching altitude either due to a tactical ATC reclearance or detection of a level bust.

It is important that the correct MCP selection is initiated in order to ensure a prompt level off at the new requested altitude. The order of selection shall be:

- ALT Hold
- Reset new MCP altitude
- Engage LVL CHG, as this will either climb or descend the aircraft to the requested altitude.

6.5. FUEL PROCEDURES

For aircraft with the new center pumps installed, no placard, the center pumps may be switched on if the center tank contains more than 460 kgs of fuel. Once the center pump low pressure lights come on the pumps should be switched off. If the captain deems that the amount of fuel in the center tank is insufficient to keep the low pressure lights extinguished during takeoff he may elect to leave the center tank pump switches off until passing FL100.

For aircraft with the old center pumps installed, placarded, the center pumps should not be switched on for T/O with less than 2,300kgs of fuel in the center tank. Once through FL100, the center pumps should be switched on. When the fuel decreases to 950kgs the crossfeed valve should be opened and the RH pump switched off. At the first indication of a center tank fuel low pressure light, the pump should be switched off and the crossfeed valve closed. In the descent when there is less than 1400kgs in the center tank, the center tank pumps shall be switched off.

6.6. FUEL CROSSFEEDING PROCEDURE

When fuel crossfeeding is required, refer to Supplementary Procedures chapter 12, Vol. 1.

Crossfeeding policy:

Prior to any crossfeeding operations, it is important to determine that a fuel leak does not exist.

It is imperative to observe fuel flow, fuel used and total fuel quantity indications prior to turning off any pump switches. Caution must be exercised when using the “LOW SIDE PUMPS OFF” rule-of-thumb.

When in climb or descent - fuel crossfeeding should be accomplished if the IMBAL alert is displayed.

Cruise - The PF will instruct the PM to accomplish crossfeed procedure as appropriate. The PM will ensure correct operation of the crossfeed valve prior to turning off any fuel pump switches. When approximately 20 kgs. of imbalance remains, the PF will instruct the PM to reconfigure for normal operations. Prior to closing the crossfeed valve, it is important to ensure the LOW PRESSURE lights are extinguished.

6.7. TRANSPONDER USE

The transponder code may be changed without selecting STBY, but consideration should be given to the entry sequence to avoid emergency codes.

7. DESCENT

7.1. INITIAL DESCENT PROCEDURES (TOD – FL150)

Flight Crew shall avoid operating on or listening to any frequency not required for current operations from top of descent to engine shut-down on stand. The intention is that pilots are giving full attention to the active ATC frequency.

Descents are normally flown in VNAV. An effective technique when descending from high altitude using VNAV PATH is to select ALT INTV 5nm prior to TOD. This enables a gentle transition from CRZ to ECON DESC.

The flight crew must verify the correct terminal procedure has been loaded.

For the Descent, the PF should select the DESCENT page on their CDU and the PM should select the LEGS page. During the approach the PF should select the APPROACH REF page. Other pages may be selected temporarily, but one should revert to the above.

Flight crews shall reduce rate of descent to 2000 FPM at 2000 feet to level-off and 1000 FPM at 1000 feet to level-off, unless a restriction set by ATC or procedure exists. This restriction shall be observed below FL200 when operating in busy TMA's or high density traffic areas.

In order to give crews a standardized format to help structure their approach setup and briefing the use of the following acronym, D A L T A, is required. Note: this structure should be used as part any approach, setup and briefing including non-normal situations and following a go-around.

D – Descent

A – Approach

L – Land

T – Taxi

A – Apron

Descent:

- Select DESC page
- FMC DESC page - Insert 250/100 and check the altitude restriction at CDU R1
- FMC Forecast page - Insert any 3 relevant forecast winds (ideally two intermediate descent winds and FL100.) QNH and ISA corrections and TAI ON/OFF altitudes where appropriate. NB, during the descent, selecting DIRECT TO an appropriate waypoint updates the FMC wind data with the current wind the point of selection.
- Verify the S.H. from the OFP, the MEA/MSA from the STAR & Approach plates and Airspace classifications from the Airfield Brief
- Any Direct Routings, planned or received from ATC, must be checked for Terrain Clearance against the most relevant source.

Approach:

- FMC DEP/ARR - Select the expected Arrival and Approach procedure. Modify the STAR if necessary to ensure that the most likely track to be flown is active
- For Procedural Arrivals, select the appropriate Approach Transition
- FMC FIX - Line select the RWxx point in the LEGS page and insert into the FIX INFO page. Where the RWxx point is not available on the LEGS page, manually type RWxx into the FIX INFO page
- FMC FIX - Insert a 3 x altitude ring, 10nm ring and a 4 or 5nm ring for NPA's depending on the Landing Gate
- FMC LEGS - Identify each separate waypoint by referring first to the arrival (STAR) plate and then to the CDU. Read across the CDU LEGS page line by line to include lateral profile, tracks and distances between waypoints and vertical constraints included in the Standard Instrument Arrival. Check for reasonableness on the ND in MAP or PLN mode. Confirm speed/altitude restrictions and Go-Around. Check the GP angle
- When planning a Circle to Land procedure, a 4.2nm range ring must also be created using the Landing RWxx point
- Tune appropriate approach aids. Set courses on the MCP.

Land:

- FMC PROG - Check the expected fuel to be used before landing
- Calculate the LDG weight and crosscheck against the Load Sheet for gross error
- Select the appropriate landing flap and VREF speed and complete the bug card
- Select autobrake as appropriate
- FMC INIT REF - Ensure the correct ILS is set and check the runway length
- Independently check and set approach minimums.

Taxi:

- Identify and brief the taxi route. Identify runway intersections, potential incursion hotspots and any single engine opportunities.

Apron:

- Check and locate the assigned stand number or expected parking area.

Use the above sequence to conduct your briefing. This will have the effect of keeping the brief logical and reduces the chances of critical items being omitted.

On short sectors it may be desirable to brief the airfield brief, safety heights, weather and NOTAM's on the ground before departure. The expected arrival and approach can be loaded in the FMC.

Descent Profile Management

Once established in the descent, frequently check if you are on profile or not. If high, above FL100, then increase speed using SPD INTV and change the FMC descent speed. This will generate a new and more efficient path based on current altitude. If high below FL 100, slow down and increase drag as necessary and conform with recommended speeds of 220 at the IAF, 180 on base, 180 on final and 160 by 4nm.

Above FL100 extra drag can readily be obtained by increasing speed. A 'mental model' of track miles to go versus altitude will highlight any possible need for additional track miles. An orbit or dog-leg will often be a solution to a possible rushed approach, especially if performed more than 10 miles from the airport at or above 3000 feet AGL. A mental cross-check at 30(FL100?) track miles to go may make the difference between an early decision to request extra track miles, or a rush and subsequent go-around. At lower altitudes the recommended modes to achieve a CDA are VNAV (assuming it has been correctly programmed in the first place and not 'ruined' by substantial shortcuts) or V/S combined with observation of the green altitude range arc. It may be difficult to achieve a CDA in LVL CHG unless speed remains constant.

Landing Performance should be calculated as normal. Where possible, the descent checklist should be completed before the descent begins.

MTRS should only be selected when operating in countries requiring altitude readouts in meters. Use of MTRS can clutter the altitude page on the PFD and which can cause problems especially on approach.

7.2. DESCENT PROCEDURES FL150 AND BELOW

PILOT FLYING	PILOT MONITORING
	Position FASTEN BELTS switch ON when passing through 15,000 feet.
Response: Call the exact passing level/altitude. "PASSING__DESCENDING__, TEN CHECKS."	Challenge: Call "ALTIMETERS" when passing FL100/10,000 feet. Ten checks are performed aloud as follows: LIGHTS: position fixed landing, turnoff and logo lights on. APU: check status FASTEN BELTS: check seat belt sign ON Check air conditioning and pressurization RECALL: check BANK ANGLE: check 25

7.3. BELOW FL100

When operating in Class A, B or C airspace, crews are restricted to 250kts below FL100 descending unless otherwise instructed. Crews should decline ATC offers of discretionary high speed below FL100 and request standard speeds. If a higher speed than 250kts below FL100 is instructed by ATC, crews shall fly this speed as instructed.

Having been given specific clearance to fly at "High Speed" below FL100, that speed should be entered into the FMC Descent page and the VNAV Path constructed around it. In this case it is acceptable to change the FMC Speed restriction from 250/100 to 250/60. These measures allow continued use of the VNAV profile while retaining its protection.

Altimeters

When cleared to an altitude, QNH should be set

Approach Checks

The change from STD to QNH is the trigger to complete the Approach checks. The acronym FISC is used to guide a pilot through the necessary features of the Approach Checks:

- Frequencies set for the approach – including the ADF frequency if relevant.
- Idents (Aural Idents of the Nav Aid are acceptable).
- Standby instruments – checked.
- Courses.

For all vectored ILS approaches and instrument approaches, using VNAV Continuous Descent Approach (CDA) for a Low Drag Approach (LDA) is the preferred descent profile to be flown. This is especially true for all approaches in the UK where our profiles are monitored. We do not do CDA approaches for any V/S NPA's. Nor do we practice V/S NPA's at any of our bases. VNAV NPA can be a CDA and LDA to 4/5nm

Selecting anything other than Auto or Normal on the Display Source panel is not authorized for normal operations.

8. APPROACH

8.1. APPROACH PROCEDURE

For the Descent, the PF should select the DESCENT page on their CDU and the PM should select the LEGS page. During the approach the PF should select the APPROACH REF page. Other pages may be selected temporarily, but one should revert to the above.

When cleared to an altitude and no approach delay is anticipated, the following sequence of preparing for the approach will be observed.

PILOT FLYING	PILOT MONITORING
Set and crosscheck altimeters at transition level or when cleared to an altitude. If PF has not set QNH by transition level, the PM reverts to the standard callout "ALTIMETERS." Both crewmembers shall then set QNH.	
"QNH____SET, PASSING____ DESCENDING ___FEET, ___ FLAGS, STANDBY ALTIMETER SET."	
Scan using the following memory aid: F - Frequencies (ADF and VOR) I - Idents S - Standby instruments C - Courses.	
Call "APPROACH CHECKLIST."	Accomplish the APPROACH checklist.
Call for flap extension according to flap speed schedule.	Position FLAP lever as directed and monitor flap and slat extension.
Approaching selected MCP altitude, verify level off and mode annunciation.	

8.2. STABILISED APPROACHES

A stabilised approach, by Virtual Ryanair definition:

- be at the correct final approach speed or correcting if less than VREF or greater than Vref +20
- the vertical speed is proportional to the current ground speed
- V/S is not more than 1000fpm continuously unless required by the published procedure and briefed prior
- on profile and in the landing configuration
- engine N1 at appropriate thrust settings

VNAV and V/S NPA requires LG Down and Flap 15 selection at 5(IMC)/4(VMC) nms, thereafter configure to landing configuration to be stable at 1000ft AAL (White Landing Altitude Reference bar) in IMC and 500ft AAL (Amber Landing Altitude Reference bar) in VMC.

During all non-precision approaches the point at which the landing configuration selections commence (i.e., gear down/flaps 15) may be modified and briefed at the captain's discretion.

It should be noted that the wind that we use for landing calculation is that reported by ATIS or the tower, not the wind in the PROGRESS page of the FMC on approach.

To calculate the fly speeds use half the headwind component plus the entire gust.

To calculate the Crosswind component use the entire calculated crosswind component from the Preamble plus half the gust.

The maximum approach speed should not exceed VREF + 20 knots or landing flap placard speed minus 5 knots, whichever is lower.

8.3. THE LANDING GATE

The aircraft shall be stabilised by the Landing Gate.

The Landing checklist shall be completed to "Landing lights."

If not, a Go-Around is mandatory.

The Virtual Ryanair Landing Gate is defined as:

Precision Approach/Non-Precision Approach	IMC	1000 feet AAL
Precision Approach/Non-Precision Approach	VMC	500 feet AAL
Visual Approach	VMC	500 feet AAL
Circling Approach or Sidestep		300 feet AAL

8.4. FLAP POLICY

Ideally using flaps as speedbrake is not recommended. However, there are circumstances when flaps AND speedbrake will be used in the course of normal operations. When executing a CDA it will often be necessary to "go down and slow down". Combining Flaps 1, 2, 5 or 10 with speedbrake can assist in deceleration and at the same time give a useful rate of descent. The configuration of Flaps 5, Speedbrake at Flight Detent and a speed of 220kts is an effective initial speed/configuration mix. To assist further deceleration use 180kts, flaps 10, and Speedbrake to Flight Detent, if necessary. This will give the best rate of descent per nautical mile.

Ideally, the use of Flaps 25 or Flaps 30 as an intermediary setting when landing Flaps 40 is desirable. It is best procedure to select the next flap setting when within 10 knots of the minimum speed for the existing flap configuration.

- Plan for Flaps 5 or more and 180 knots or less at glideslope interception. This is correct for all final approach paths.
- Ideally the following procedures are used for flap extension:
  - Select flaps 1 when decelerating through the flaps–up maneuvering speed, displayed on the airspeed display as a "UP.
  - Set airspeed cursor to the flap maneuvering speed displayed as "1."
  - When appropriate, select the next flap position and then set the airspeed cursor to that flap maneuver speed.
  - The PM response "Speed checks, flaps__", requires the PM to ensure that the current speed is not in excess of the limit speed for the flap setting. It also allows the PF an opportunity to correct a request for an erroneous flap setting and to confirm that the PM is selecting the requested flaps setting.

Note: Flap maneuver speeds provide approximately 15 to 20 knots above the minimum maneuvering speed for each flap setting.

When on final approach in landing configuration, it is not recommended to set the A/T command speed to allow for wind or gust corrections unless a manual landing is anticipated. Through airspeed and acceleration sensing, the A/T corrects for normal wind gusts. Higher command speed settings result in excessive approach speeds. The recommended A/T approach speed setting is VREF + 5.

The maximum approach speed should not exceed VREF + 20 knots or landing flap placard speed minus 5 knots, whichever is lower.

8.5. FLAP EXTENSION SCHEDULE

Current Flap Position	At Speedtape "Display"	Select Flaps	Command Speed for Selected Flaps
UP	"UP"	1	"1"
1	"1"	5	5
5	"5"	15	15
15	"15"	30 or 40	(VREF30 or VREF40) + wind additives

Ideally the use of flaps 25 or flaps 30 as an intermediary setting when landing flaps 40 is recommended. Use flaps 2 and flaps 10 as required to comply with speed restrictions.

8.6. AUTOBRAKE, REVERSE THRUST AND FLAP

All Landings, a Serviceable Aircraft, No Applicable NNCs.

The PF, with the agreement of the Commander shall decide on the Landing Flap, Autobrake and Reverse Thrust settings having consulted the QRH, PI section. When an OFP note or Airfield Brief specifies a particular landing configuration and/or technique these configurations/settings shall be used.

A minimum of Idle Reverse Thrust shall be used for all landings along with Autobrake 0-1-2-3-Max as appropriate. Where the use of higher Autobrake brake settings causes turnaround brake cooling issues, Reverse Thrust settings greater than idle shall be used.

Idle Reverse Thrust therefore shall be used in normal flight operations with braking action GOOD or better as follows:

- On last flight of the day at all Virtual Ryanair bases.
- On the last flight of a series of flights when there is at least a 60 minute turnaround.
- Last landing prior to aircraft going out of service.
- Any landing when the turnaround schedule will not be adversely affected by the sole use of Autobrake.

The Commander is encouraged to use a combination of Flap/REV THR/AUTO BRK in order to exit the runway at a particular exit to minimize the taxi time to stand.

Note: The operational objective is to minimize runway occupancy times consistent with safety of operations.

Safety

Our primary operations objective is to conduct our air transport activities safely. Use of idle reverse thrust as described in this section is not considered detrimental to safety.

When required, up to and including MAX REV THR may be used.

Reverse Thrust Detents and Max Reverse

First detent reverse thrust is IDLE REV. Second detent reverse thrust is approx. 75% N1. MAX REV THR is approx. 82% N1.

8.7. VISUAL APPROACH

Definition:

An approach where either part of, or all of an instrument approach procedure is not completed and the approach is executed with visual reference to the terrain.

Concept:

A visual approach in Virtual Ryanair is normally carried out because there is no approach procedure for that runway or a time saving can be achieved by carrying out a visual approach as opposed to a long protracted arrival and approach procedure.

Weather Conditions:

The runway must be in sight and remain in sight during the visual approach.

General:

Self maneuvering night time visual approaches are prohibited.

Except where there is an existing FMC approach for the runway, the pilot should generate an RX point at 4.0nm from the runway threshold with a 3 degree glide path (or PAPI glide path angle, if available.) The corresponding altitude at this point should be made a “hard altitude” in the FMC. The pilot should use LNAV track and VNAV path information for guidance. When cleared for the approach and terrain clearance is guaranteed, select ‘direct to’ the RX point. Once the “Direct To’ has been selected, the PF shall review the path and shall not turn final unless on path +/-400’.

Crews shall plan to be established on final approach with the landing gear down and flaps 15 by 4nm from the runway.

The landing gate for a visual approach shall be Land Alt plus 500 feet, which shall be set on the BARO MINS. The EGPWS, “FIVE HUNDRED” callout will act as a minimums call.

The pilot should maximize the use of the AFDS. The autothrottle should remain engaged until at least 4nm in order to ensure that low or excessive speeds are avoided during the manoeuvre. The autopilot, if in use, should be disengaged before minimums.

Visual Approach with an Existing FMC Straight-In procedural Approach for the runway:

The crew can adapt any FMC arrival in order to give them lateral and vertical guidance for the visual approach. This will also ensure that any missed approach procedure available in the FMC can be used in case of a go-around.

The PF shall identify and agree to a point on the approach that equates to not less than 4nm from the runway and plan to be established at or before this point with the landing gear down and flaps 15 extended. Once the ‘Direct To’ has been selected to this point the PF shall review the path and shall not turn final unless on path +/- 400’.

Other than in the circumstances described above, crews should use the runway extension option in the FMC.

Other Visual Approaches:

- Without an existing Procedural Approach for the runway
- With an offset existing Procedural Approach for the runway.

Crews must exercise caution before accepting a visual approach to either of the above types of runway.

If no procedural approach exists, there could be a terrain issue which precludes the use of normal PANS OPS criteria for constructing an approach. This could make that runway unsuitable for a straight in visual approach. If any doubt about terrain clearance exists, the visual approach must be refused.

A significant offset to the front course of a published approach will almost always be the result of a terrain issue that affects the straight-in track. The commander must decide whether the offset and the reason for it allow complete clearance from terrain during any visual approach. If any doubt about terrain clearance exists, the visual approach must be refused.

Go-around instructions must be agreed with ATC (if none are published) and followed using TOGA, HDG SEL, VOR/LOC and MCP altitude.

8.8. TIMED APPROACHES

Timed non-precision approaches are approved for Virtual Ryanair Operations.

All Virtual Ryanair aircraft shall observe the maximum ICAO recommended speeds of 220 KIAS at the IAF and 180KIAS on inbound turn (particularly important.)

1. Cross the IAF with F1 (220kts max.)
2. Proceed outbound as per published timing.
3. Commence the inbound turn F5 (180kts max.)
4. Select gear down and F15 and do the landing checks. (5 mile ring when using VNAV or established on inbound QDM when using V/S.)
5. F30/40 and descend to MDA (max V/S 1000/miin.)
6. At MDA, if unable to continue to a successful landing, initiate a standard missed approach procedure.

If the approach is available in the FMC Nav database:

(a) LNAV may be used as the magenta line takes into account the timing limitations of the outbound leg.

(b) VNAV may be used subject to: An appropriate path with the MAP at or before the runway threshold, and/or a glidepath is published on the LEGS page.

Note: The Double Brief covers all operational issues and the use of the 5 mile ring from the RWxx point identifies the configuration point.

Note: Use published minimums adjusted as per normal SOP.

If the procedure is not available in the FMC Nav database:

(a) Use HDG SEL + V/S

(b) The MDA shall be the published MDA + 300’ or 1000’ AAL, whichever is higher.

(c) Once established on the inbound QDM, gear down F15 LDG checks.

Note: For V/S NPA’s use the published GS/ROD guidance in the approach chart.

A three degree ROD can be calculated by GS x 5 e.g. 160kts x 5 = 800fpm.

8.9. SURVEILLANCE RADAR APPROACHES, SRA

Definition:

An SRA is a Non Precision Approach (NPA) where the radar controller gives heading instructions to position the aircraft on final approach. The controller will instruct the crew to commence descent.

During final approach the controller will give the crew their distance to touchdown and advice the altitude they should be at.

An SRA is usually terminated at a range of 2NM from the runway threshold but could terminate as close as 0.5NM from threshold.

General:

An SRA cannot be conducted unless the Virtual Ryanair "Double Brief" has been completed.

Where the missed approach procedure is not included on the SRA Chart, the missed approach procedure including stop altitude must be verified with ATC prior to commencement of the approach.

A missed approach must be executed immediately in the event of a radio failure.

Establish the descent point. In the FIX Page enter a 10 NM ring and a ring that coincides with the expected Descent Point.

If an approach with identical missed approach is available in the FMC select it for missed approach guidance. If not create an RX point at the descent point with the published glide path angle for display on the ND.

Set revised MDA on the BARO MINS. Use HDG SEL and V/S. Aim to leave the platform altitude in landing configuration.

PM calls out advisory altitudes Vs distance as required.

The controller will expect verification of gear down and locked on final.

PF may call visual at any time when sufficient visual cues are established. The controller will then terminate guidance.

Procedure:

1. PM calls "Approaching Descent" 2 NM before Descent Point
2. PF confirms ALT HOLD and calls "XXX feet set" (Revised MDA)
3. Select Gear Down and flaps 15 at 1 NM before Descent Point
4. Select Landing Flaps 0.5 NM before Descent Point
5. Commence Descent when instructed by ATC
6. At MDA, if unable to continue to a successful landing, initiate a standard missed approach procedure

8.10. 500 CONTINUE/500 GO-AROUND CALL

In order to make the "500 Continue" call, the PM must have established the following:

- Speed - VREF to VREF + 20 knots
- Vertically - on glidepath (+/- 1 dot or 3 reds or 3 whites)
- Laterally - on centerline (+/- 1 dot)
- Appropriate thrust set
- Landing checklist completed (except landing lights).
- and the vertical speed is proportional to the current ground speed, but not more than 1000fpm unless briefed prior

If any of these parameters cannot be confirmed, the call shall be "500 GO-AROUND"

The "500" call on approach is a vital safety tool. Pilot must react to a 500 "GOAROUND" call as they would to an autoland "Go-Around" call. There should be no discussion at this point just an immediate go-around.

Corrections to maintain a stabilized approach are permitted inside the landing gate. However, any FLAP BLOWBACK, TOO LOW GEAR/FLAP or a TERRAIN caution at night or IMC requires an immediate go-around.

The Landing checklist is considered complete at the LDG Lights, from a LDG Gate point of view.
If the PF is flying manually, then the PM will switch the LDG lights on once cleared to Land.

The Senior Virtual Ryanair Staff require a go-around be executed if a crew receives an EGPWS hard warning below the Landing Gate.

8.11. LANDING PROCEDURE AUTOLAND

PILOT FLYING	PILOT MONITORING
When on localizer intercept heading and within ILS Designated Operational coverage (DOC), verify the ILS is tuned and identified, LOC and G/S pointer are correctly displayed, arm the VOR/LOC mode when cleared to establish on the localizer	
Verify mode annunciation.	
At localizer capture verify proper mode annunciation and call “SET RUNWAY HEADING ____ °” or “RUNWAY HEADING ____° SET.”	Verify proper mode annunciation.
When cleared for the approach, arm the APP mode and engage the second autopilot.	Verify proper mode annunciation.
At 5 NM, call “GEAR DOWN”, “FLAPS 15." Call “LANDING CHECKLIST DOWN TO FLAPS.”	Position landing gear lever DN, FLAP lever to the 15 detent. Position engine start switches to CONT. Arm speed brake. Check RECALL.
	Accomplish the LANDING checklist down to flaps. Call “HOLDING AT FLAPS.”
At glide slope capture, verify proper mode annunciation, check N1 reference bug at the go– around limit and set missed approach altitude.	
“XXXX FEET SET.”	“GLIDESLOPE CAPTURE.”
The PF shall point to the Missed Approach Altitude on the FMC LEGS page and call “XXXX FEET.” The PF shall then set that Missed Approach Altitude in the MCP Altitude, point to it and call “SET.”	
Call “FLAPS 30/40” as required for landing. Set ____ knots.	Position FLAP lever as directed. Set speed.
Call “COMPLETE THE LANDING CHECKLIST.”	Complete the LANDING checklist.
Call: “OM___FEET, FLAGS/NO FLAGS.”	Call: ALT CHECKS, FLAGS/NO FLAGS.”
Monitor autopilots as required.	Monitor approach progress.

Call: “PASSING ___ RADIO, FLARE ARMED.”	At 500 feet RA call: “500 RADIO FLARE ARMED.”
Call: “CHECK.”	GPWS “PLUS HUNDRED.” The captain starts looking for visual cues.
GPWS “MINIMUMS.”	Call: “LAND or GO–AROUND.” If “LAND”, assume control and monitor flight path to touchdown. Select reverse thrust and disengage the A/P after nosewheel touchdown.
If the captain calls “GO-AROUND” push TOGA and call “GO-AROUND, FLAPS 15.” If the call is “LAND,” relinquish control and monitor the flight instruments and systems throughout the landing. Bring failures to the attention of the captain.	Call: “LAND or GO–AROUND.” If “LAND,” assume control and monitor flight path to touchdown, select reverse thrust and disengage the A/P after nosewheel touchdown.
At approximately 50 feet AGL, verify FLARE is engaged. Ensure the autothrottle retards the thrust levers to idle by touchdown. The first officer will remain on instruments until taxi speed.	

8.12. LANDING PROCEDURE SINGLE CHANEL OR MANUAL

PILOT FLYING	PILOT MONITORING
When on localizer intercept heading and within ILS Designated Operational Coverage (DOC), verify the ILS is tuned and identified, LOC and G/S pointer are correctly displayed. Arm the APP mode when cleared for the ILS.	
Verify mode annunciation.	
At localizer capture verify proper mode annunciation and call "SET RUNWAY HEADING ____ °" or "RUNWAY HEADING ____° SET."	Verify proper mode annunciation. Select appropriate heading, if required.
At 4nm, but not later than 3.5nm, call "GEAR DOWN", "FLAPS 15." Arm speed brake and check green light illuminated. Call "LANDING CHECKLIST DOWN TO FLAPS."	Position landing gear lever DN, FLAP lever to the 15 detent. Position engine start switches to CONT. Check RECALL.
	Accomplish the LANDING checklist down to flaps. State "HOLDING AT FLAPS."
At glide slope capture, verify proper mode annunciation, check N1 reference bug at the go– around limit and set missed approach altitude	
"XXXX FEET SET."	"GLIDESLOPE CAPTURE."
The PF shall point to the Missed Approach Altitude on the FMC LEGS page and call "XXXX FEET." The PF shall then set that Missed Approach Altitude in the MCP Altitude, point to it and call "SET."	
Call "FLAPS ____" as required for landing. Call: "____ KNOTS" or "SET ____ KNOTS."	Position FLAP lever as directed. Set speed.
Call "COMPLETE THE LANDING CHECKLIST."	Complete the LANDING checklist.
Call: "OM ___ FEET, FLAGS/NO FLAGS."	Call: ALT CHECKS, FLAGS/NO FLAGS."
Monitor autopilot as required.	Monitor approach progress.
	At 500 feet to touchdown call: "500 CONTINUE" or "500 GO-AROUND".

Call: “CHECK.”	GPWS “PLUS HUNDRED”
Call: “LAND” or push the TO/GA switch and call “GO–AROUND, FLAPS 15.”	When sufficient visual reference stablished, at or before minimums, call “RUNWAY/ LIGHTS IN SIGHT /AHEAD/LEFT/RIGHT.” GPWS “MINIMUMS” Call: “GO-AROUND” if visual reference not achieved.

The go-around must be initiated by DA. Therefore, at the minimums call, it is important that the PF, without delay, accomplishes the go-around. It is not necessary for the PF to search for visual cues at this point. The PM shall call the visual cues as they become apparent. If the PF has not heard any visual cue calls from the PM, it is safe to assume that no visual cues have been received. Execute a go-around.

8.12.1. ILS APPROACH – GUIDANCE

Where possible a CDA and low drag type approach shall be accomplished. This provides the best noise footprint and lowest fuel burn. Careful construction of the descent profile for CDA approaches is required. It is best practice to continually update the\VNAV descent profile by going “Direct To” a convenient waypoint on the approach. When this is no longer practicable, the centre line may be extended. V/S should then be used in order to finesse G/S capture.

Aim to be within 10 knots of the Up speed before selecting Flaps 1 and the maneuvering speed for each other configuration change. On G/S interception, maintain Flaps 5/10 as required in order to maintain correct speed. Passing 4 DME select landing gear down, Flaps 15 and landing check before 3.5 DME.

8.13. LOW VISIBILITY APPROACHES - GUIDANCE

Virtual Ryanair is approved to operate to minimum CAT II and CAT III minima of RVR 300m/100 ft RA and RVR 200m/50 ft RA respectively. The minimum requirements for the mid-point is 125m and 75m for stop end.

Additionally the runway must be 45m or greater with cleared width of 30 m or greater to conduct autoland. ("Blacktop" of 30m cleared width and/or 30m runway width is acceptable for non autoland operations). The crosswind limits should be calculated for 45m and reduced by 1 kt/metre reduced/cleared width. Refer also to Table 1 & 2 "Downgraded Equipment, Effect on landing Minima" in OM A.

Always remember there are 4 elements to Autoland:

- Crew "Qualified"
- Aircraft "Qualified"
- ILS "Qualified"
- Runway "Qualified"

Operations Manual Part A Ch. 8 give the option of disregarding the stop-end RVR if the aircraft can stop within the first 2/3rd of the runway. Use the pre-amble to find out if this is possible.

Downgraded lighting systems will have an effect on the RVR minima used. The information required is in Operations Manual Part A, Ch. 8 and this must be studied by the crew, if required.

Gross error check: When passing 1500 RA the G/S and LOC pointer will flash during the BITE test, after which FLARE arm will be annunciated. Failure of FLARE to arm is most likely caused by omission of CMD A selection and this can still be rectified if noticed immediately after the BITE test and above 800ft RA.

8.14. NON PRECISION APPROACHES (NPA)

All non precision approaches are challenging procedures that contribute to aircraft accident and incidents to a far greater extent than a precision approach. Virtual Ryanair's DALTA and Double Brief procedures are designed to alleviate some of the latent threats associated with non precision approaches and must be followed strictly.

Despite the fact that aspects of our instrument approach using VNAV procedure resembles an ILS (e.g. VNAV path indicator) it is still a non precision approach and all land based Nav Aids must be tuned, identified, active, and referenced to by flight crew.

All non precision approaches are prone to the threat of QNH Blunder Error (I.e. using an incorrect QNH). If this error has been made no distance/altitude crosscheck will reveal that the aircraft is in a potentially catastrophic situation that will only become apparent with an EGPWS Caution or Warning. Accordingly it is essential that:

Crews carefully crosscheck the ATIS or ATC supplied QNH against the METAR QNH value that should have been recorded on the loadsheet.

Know precisely the terrain escape maneuver and use it immediately if a Terrain Warning is announced by the EGPWS.

8.15. INTRUMENT APPROACHES USING VNAV

PILOT FLYING	PILOT MONITORING
Enter the RWxx (runway) waypoint in the Fix info page. Enter a 10 nm ring and either a 4 nm ring (VMC) or a 5 nm ring (IMC) around the runway.	Notify the cabin to prepare for landing. Verify that the cabin is secure.
Extend flaps 1 prior to the 10nm ring. Call "FLAPS___" according to the flap extension schedule. Select SPD INTV if VNAV engaged.	Set the flaps lever as directed. Monitor flaps and slats extension.
The recommended roll modes for the final approach are: For a VOR or NDB approach use LNAV For a LOC, approach use VOR/LOC	
One mile before set all charted intermediate altitudes between the IAF and Descent Point on the MCP altitude selector.	
When on a final approach course intercept heading for LOC approaches: Verify that the localizer is tuned and identified Verify that the LOC pointer is shown Verify that the FMC Descent Point coincides with published distances	
Select LNAV or arm the VOR/LOC mode	
WARNING: When using LNAV to intercept the localizer, LNAV might parallel the localizer without capturing it. The airplane can then descend on the VNAV path with the localizer not captured.	
Use LNAV or HDG SEL to intercept the final approach course as needed.	
Verify that LNAV is engaged or that VOR/LOC is captured	
2 NM before the ND Descent Point: Set MDA on the MCP Select or verify VNAV PATH Select or verify speed intervention	Call "APPROACHING DESCENT"

At the Final Approach Fix, verify the crossing altitude and crosscheck the altimeters.	
Monitor the approach, including distance/altitude checks	
At 4 or 5 nm (VMC/IMC) to the runway call: “GEAR DOWN” “FLAPS 15” “LANDING CHECKLIST TO FLAPS”	Set the landing gear lever to DN. Verify that the green landing gear indicator lights are illuminated. Set the flap lever to 15. Set the engine start switches to CONT
Set the speed brake lever to ARM. Verify that the SPEED BRAKE ARMED light is illuminated.	Do the landing checklist down to flaps.
Call “FLAPS___” as needed for landing.	Set the flap lever as directed.
Call “COMPLETE LANDING CHECKLIST”	Do the landing checklist.
1000 feet AAL The PF shall point to the Missed Approach Altitude on the FMC LEGS page and call “XXXX FEET.” The PF shall then set that Missed Approach Altitude in the MCP Altitude, point to it and call “SET.”	
If suitable visual reference is established at MDA/DA or the missed approach point, disengage the autopilot and autothrottle. Maintain the visual glide path to landing.	Recycle flight directors

8.16. NON-PRECISION APPROACHES USING VNAV

The MDA/DA is set on the PFD as published on the approach chart (subject to temperature correction). The MCP altitude is set to nearest 100ft above the MDA/DA.

The briefing and FMC setup are essential to a successful VNAV approach. Situational awareness in relation to the vertical path is very important especially during the transition to the approach, i.e. when on radar vectors or during the arrival procedure. Stay on or slightly below the path at all times in order to avoid excessive pitch down when engaging VNAV.

Fly-offs are created when the FMC programmed centrefix is located before the Jeppesen published GP, and an “at” altitude is programmed at the centrefix. In this case use the centrefix as the DP. Aim to be at platform altitude at the centrefix. The aircraft will then fly a level segment to the T/D point, where it will start descending towards the MDA (MDA will have been set on the MCP 2 nm before the centrefix).

Slow down at the Decel point or 10 nm if earlier or if no Decel point is displayed on the ND. Select flaps and SPD INTV at the up speed.

Ideally VNAV should be used as early as possible during the descent or arrival procedure. When on radar vectors for a VOR or NDB approach use the following sequence for MCP selections:

“Cleared to Establish on the Front Course” - Select LNAV

“Cleared to Descend with the Procedure” - Select VNAV (if not already selected)

During radar vectors for a localizer approach proceed as above but use VOR/LOC instead of LNAV. In all cases VNAV PATH mode must be engaged at 2 NM prior to the Descent Point (DP).

Monitor raw data for confirmation that the approach front course has been correctly intercepted and maintained.

When in VNAV select speed intervention after selecting flaps. The thrust mode will now be FMC SPD and the pitch mode VNAV PTH.

The ND distance readout in the upper right corner is the primary distance readout for the Descent Point. At 2nm to the DP set the MCP altitude to the nearest 100ft above the MDA/DA, engage or verify VNAV PTH, and SPD INTV.

The ANP must be monitored during the VNAV NPA. The only indication of the FMC transitioning into the “ON APPROACH” logic is when the RNP changes to 0.5 on the legs page.

If RNP 0.5 is not displayed at this stage the approach should be continued, the “ON APPROACH” logic will become active at the latest when descending through 2000 feet AAL.

There is no below path alerting like there is for ILS approaches. Therefore the A/P shall be used whenever it is available to reduce the risk of deviating from the path.

Sufficient Altitude/Distance crosschecks on the Jeppesen chart must be readout by the PM and cross checked by PF during the approach to ensure that the correct path is being maintained. It is not essential that each and every Altitude/Distance crosscheck is called especially if this is impacting on the efficient management of the cockpit.

Landing gear must be selected at 5nm from the RW point in IMC, and 4nm from the RW point in VMC at the latest. Landing Flap shall be selected to ensure that the Landing Gate is not penetrated. With high MDA/DAs or in strong tailwind conditions the Captain may choose to extend the landing gear and associated flap at an earlier point in the approach.

Set the MAA at 1000' AAL (White altitude reference bar on altimeter). If the MDA/DA is 900' AAL or higher, then the MAA must be set earlier to avoid ALT ACQ. This point must be agreed by the flight crew prior to the approach. In the unlikely event that the MAA is below 1300' AAL then the MAA must be set 300' below MAA.

If suitable visual reference is not established at MDA/DA then a go-around must be conducted.

When in VNAV PTH, without SPD INTV selected, due to system design a "THRUST REQUIRED" message will appear if the speed goes 5 knots below the UP speed. However the A/T will not add thrust until 15 knots below the UP speed. The PF remains responsible for adding drag or thrust as required to manage the speed and energy of the aircraft. Do not fly at speeds below maneuvering speeds.

8.17. CIRCLE TO LAND APPROACH

During the set-up for this approach ensure that a 4nm circle is established in the FIX page from the RW point of the Instrument runway. It is reasonable to expect that most circle to land approaches will be flown in "VMC" which will allow a 4nm reconfiguration point. If necessary use a 5nm ring as per NPA SOPs.

In addition, a 4.2nm ring must be established around the RW point for the landing runway. The 4nm and 4.2nm rings will display a very accurate depiction of the PANS OPS protected area for circle to land approaches.

On the PFD set revised MDA/DA (circling MDA/DA rounded up to nearest 100 feet)

The instrument approach that results in a circle to land shall normally be flown using the following AFDS modes:

- Roll - LNAV for VOR and NDB approaches. HDG for approaches not contained in the FMC database. LOC for a localizer approach.
- Pitch - V/S or VNAV

With the circling MDA set in the MCP, the AFDS pitch FMA will annunciate ALT ACQ and ALT HLD as minimums are approached. During this sequence PF shall proceed as follows:

- ALT ACQ
  - Set the heading associated with the track which is offset at 45 degrees to the runway.
- At ALT HLD
  - Select HDG SEL at MCP
  - Set MAA - the PF shall point to the Missed Approach Altitude on the FMC Legs page and call "XXXX FEET." The PF shall then set the Missed Approach Altitude in the MCP, point to it and call "SET."

Timing is started when steady on track or wings approximately level whichever is sooner. Both pilots will time the leg. PM selects PROG page 2 at this point. Wind effect must be factored into the duration of this leg by reducing/increasing the still air time of 20 seconds by an amount that is equal to half the TWC/HWC. TWC/HWC can be seen in the PROG page 2 in the FMC. HDG is used to ensure that the required track is maintained. At Time Out turn onto the downwind track using HDG to adjust for wind.

On the ND select 5nm range, with this setting, the aircraft symbol should be just outside the inner range arc at a distance of approximately 1.25nm. Any less than this and the aircraft is too close to the runway. Any more than this and the aircraft is too far away.

Keep the runway in sight at all times. Abeam the landing threshold start timing. The timing required is 3 times the HEIGHT above the runway less half the TWC in seconds. Do not forget to include the amount that has been rounded up to the revised MDA when making this calculation.

When TIME OUT is noted proceed as follows:

Keep the landing threshold and PAPI's in sight

- Turn the MCP HDG to the runway heading
- Call for landing flaps
- Select MCP speed to V Fly
- Descend on the PAPI's
- Complete the landing checks
- Disconnect the Autopilot at MDA - 50 feet.
- If desired, leave the Autothrottle engaged until a stable and appropriate thrust setting has been established to ensure an on-speed, on PAPI rate of descent.
- Disconnect the Autothrottle before 500ft AAL

The position trend vector is invaluable in giving early indication of how the aircraft will roll out relative to the runway extended centre line. Use the bank angle selector to assist in adjusting the rate of turn as appropriate when above the Autopilot MUH.

When disconnecting A/P and A/T, call "RECYCLE FLIGHT DIRECTORS."

The landing gate is 300 feet. However, the landing configuration must be established by 500ft AAL. The "CONTINUE/GO-AROUND" call is made at 300' AAL.

If a Go-Around from a circle to land is required, the missed approach procedure for the instrument approach runway must be used. Care must be taken to control speed and therefore radius of turn so as to remain within the 4.2nm circle around the Runway thresholds of the Instrument Runway and the Landing Runway. The initial turn must always be towards the runway and may require a turn in excess of 180 degrees.

Proceed as follows:

- Select TOGA and advance the thrust levers towards go-around thrust
- Call "GO-AROUND FLAPS 15, SET GO-AROUND THRUST" (90% N1)
- Call "GEAR UP" (with a positive rate)
- At 400 feet AGL call "HDG SEL"
- Maintain flaps 15 until established on the missed approach track

Climb straight ahead until circling minima if already on finals. At this altitude commence a turn in the direction of the downwind leg just flown to intercept and establish on the track required by the missed approach procedure for the instrument runway. The flight director roll and pitch guidance may be followed during this maneuver. The pitch bar will direct a speed of 172 knots with flaps 15 selected which will ensure that Category C performance is met.

When established on this track and above circling minimums, accelerate and retract the flaps on schedule. Re-engage the AFDS as per Virtual Ryanair SOP.

Note that the Configuration Warning may sound as the thrust levers are retarded in order to level off at the missed approach altitude.

If at ALT ACQ the MCP speed window opens, the PF shall call "Bug Up" and ensure that flap retraction is completed promptly in accordance with the correct flap retraction schedule.

8.18. NON-PRECISION APPROACH USING VS

VS NPA is a supplementary procedure that should only be used when there is no NPA or overlay approach in the FMC for the runway in use. A successful NPA is dependent on a correct briefing and flown to the Virtual Ryanair profile.

If the NPA profile is not in the FMC nav database, then LNAV may not be used for the approach. HDG SEL and VS will be the correct modes used in a VS NPA.

Plan to be level at least 2nm before the descent point in order to successfully fly NPA using V/S procedure. It is important to have a hard altitude set at this distance in the FMC e.g. 3000 (2 nm minimum) prior to the Descent Point for the approach. Inputting a waypoint 2nm before the descent point at the required altitude will greatly help your descent profile.

Flaps 1 at 10nm before the airport and approximately 3000 feet is a limit and not a target. It is good practice to be at Flaps 1 earlier. If outside these parameters then take corrective action early in order to avoid the possibility of a High Energy Approach.

When briefing the approach, nominate a distance for Flaps 1 and 5. In this way if the PF forgets to configure on schedule, the PM is in the loop to remind him.

At the Descent Point start the final approach with a VS of 1000 feet per minute. Calculate Rate of Descent by using five times Ground Speed. Use the Altitude Range Arc to ensure that the path is correct.

While the aircraft must be in gear down and flaps 15 configuration by 4nm or 5nm, the landing Gate is still 500/1000 feet. This means fully configured with all checklist items completed, except landing lights.

Set the MAA at 1000’ AAL (White altitude reference bar on altimeter). If the MDA/DA is 900’ AAL or higher, then the MAA must be set earlier to avoid ALT ACQ. This point must be agreed by the flight crew prior to the approach. In the unlikely event that the MAA is below 1300’ AAL then the MAA must be set 300’ below MAA.

If suitable visual reference is not established at MDA/DA then a go-around must be conducted.

Remember we can’t legislate for all approaches, so good Airmanship should prevail when deciding the landing gear extension point. High MDAs, steep approaches or tailwinds, for example will suggest a reconfiguration point before 5nm.

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9. GO-AROUND

Any go-around is a difficult maneuver that may not have been completed since a Recurrent Simulator session, possibly 6 months previously. It is an extremely dynamic procedure where the aircraft is climbed, accelerated and reconfigured in a very short space of time and often, airspace. As a result, it frequently leads to problems the most common of which are Level Busts and exceedences of flap limiting speeds. Note that SOPs call for a review of the go-around procedure during the approach briefing. Good Airmanship/TEM dictates that this takes the form of a very thorough description of the exact sequence of calls, actions and MCP selections which will be made, including the landing gate that may determine the continue/go-around decision. A thorough briefing will result in a greatly reduced workload during any approach and go-around. Virtual Ryanair operates a no-blame policy provided landing gates are respected.

Many FMC missed approach procedures show heading 'vectors' in the legs page and in this case it is necessary to use HDG SEL on the MCP with an appropriate wind adjusted heading.

9.1. GO-AROUND PROCEDURE DUAL CHANNEL

PILOT FLYING	PILOT MONITORING
Push TO/GA switch. Call “GO-AROUND - FLAPS 15.” If full G/A thrust is required, push the TO/GA switch again after reduced G/A thrust is established.	Confirm thrust advances toward G/A. Call “FLAPS 15”, position FLAP lever to 15 and monitor flap retraction.
Confirm rotation to go–around attitude and monitor autopilot. Call “SET GOAROUND THRUST.”	Verify correct autothrottle operation and call “GO-AROUND THRUST SET.”
Verify mode annunciation.	
When positive rate of climb is indicated, call “GEAR UP.”	Verify that both VSI and altimeter indicate a positive rate of climb and call “POSITIVE RATE” and move the gear lever to the UP position.
Check flight instrument indications (MCP speed window blanks.)	
Above 400 feet, select appropriate roll mode and commence flap retraction.	Verify mode annunciation. Position FLAP lever as directed, monitor flaps and slats retraction and call “FLAPS UP, NO LIGHTS.” Call ATC
Call “TUNE RADIOS FOR MISSED APPROACH.”	Tune radios as directed.
Verify airplane levels off at selected altitude.	
Call “AFTER TAKEOFF CHECKLIST.”	Accomplish the AFTER TAKEOFF checklist.

9.2. GO-AROUND PROCEDURE SINGLE CHANNEL OR MANUAL

PILOT FLYING	PILOT MONITORING
Push TO/GA switch. Call "GO-AROUND - FLAPS 15." If full G/A thrust is required, push the TO/GA switch again after reduced G/A thrust is established.	Confirm thrust advances toward G/A. Call "FLAPS 15", position FLAP lever to 15 and monitor flap retraction.
Rotate to go–around attitude and call "SET GO-AROUND THRUST."	
Verify mode annunciation.	
When positive rate of climb is indicated, call "GEAR UP" and monitor acceleration.	Verify that both VSI and altimeter indicate a positive rate of climb and call "POSITIVE RATE" and move the gear lever to the UP position.
Check flight instrument indications (MCP speed window blanks.)	
Above 400 feet, call for appropriate roll mode and commence flap retraction.	Verify annunciation. Position FLAP lever as directed, monitor flaps and slats retraction and call "FLAPS UP, NO LIGHTS." Call ATC.
Call "TUNE RADIOS FOR MISSED APPROACH."	Tune radios as directed.
Level off at selected altitude and maintain flaps-up maneuvering speed.	
Engage autopilot and call "COMMAND A/B."	Verify annunciation.
Call "AFTER TAKEOFF CHECKLIST."	Accomplish the AFTER TAKEOFF checklist.

9.3. GO-AROUND GUIDANCE

When manual thrust is used, the target N1 should not exceed the Go-Around N1 bugs.

When the autothrottle is engaged one push on the TOGA switch normally gives enough thrust for go-around. Verify on the FMA that GA is annunciated in the A/T channel (N1 if TOGA is pushed twice) and that 1000-2000 feet Rate of Climb is achieved on the VSI.

At 400 feet RA engage a roll mode and select flaps 5 if the speed is at or above the 15 bug. Prompt flap retraction is important.

The commanded speed (magenta bug) is determined by actual flap position and based on max take-off mass. The speed window on the MCP is closed therefore the target speeds are controlled by actual flap position. Vref 40 for 74990 is 152 knots, therefore the AFDS will command the following speeds:

- Flaps 15 – 172 knots (Vref 40 +20)
- Flaps 5 – 182 knots (Vref 40 +30)
- Flaps 1 – 202 knots (Vref 40 +50)
- Flaps Up – 222 knots (Vref 40 +70)

The PM should keep his/her hand on the flap lever in order to ensure a timely flap retraction. The PF must commence the flap retraction schedule as soon as possible after 400 feet RA.

The PF shall keep both hands on the control column until the “FLAPS UP” call is made.

10. LANDING

10.1. LANDING ROLLOUT PROCEDURE

PILOT FLYING	PILOT MONITORING
Ensure thrust levers at idle.	
Disengage autopilot and control airplane manually. Verify autothrottle disengages automatically.	Verify autothrottle is disengaged.
Verify SPEED BRAKE lever UP	Verify SPEED BRAKE lever UP. Call out "SPEEDBRAKES UP." If SPEED BRAKE lever not UP, call "SPEEDBRAKES NOT UP."
Verify proper autobrake operation.	
Without delay, raise reverse thrust levers to the interlocks, hold light pressure until release, and then apply reverse thrust as required.	Monitor engine instruments and announce any engine limit being approached, exceeded or any other abnormalities.
At 60 knots, reduce reverse thrust to be at IDLE reverse when reaching taxi speed	Call "1/2/NO REVERSER(S) ____ %" Call: "100." Call: "80." Call: "60."
Approaching taxi speed, slowly move the reverse thrust levers to the full down position.	Verify REV indication extinguished.
Prior to taxi speed, disarm the autobrake and continue manual braking as required	

The landing phase begins at 50ft. Once the MLG is on the runway, gently lower the nose and select reverse thrust as required. Do not trim during the flare or touchdown. This increases the possibility of a tailstrike during landing.

WARNING: After reverse thrust has been initiated, a full stop landing must be made.

In crosswind landings the “de-crab during flare” and “touchdown with crab” techniques are normally used. Refer to FCTM.

Runway occupancy is important. This does not mean that we make the first exit off every runway, but that we should expedite departing the runway at whatever exit is selected or allocated. Max speed to exit a RET (Rapid Exit Taxiway) is 60kts allowing for runway conditions, recommended speed is normally 45kts and will vary lower depending on the design of the specific exit which should be checked again the airfield plates/briefs. At all times, safety is paramount and passenger comfort should be considered. If the turnoff is 90 degrees the max speed is 10kts.

10.2. RUNWAY VISUAL AND LANDING

Use small thrust, pitch and roll adjustments. The most common mistakes are pitching for speed or applying thrust for height.

A fixed landing point is projected through the windscreen down to just before the 1000 foot point. That is the aim point. The idea is to fly the aircraft down to 50 feet keeping the aim point constant in the windscreen.

Some pilots that have difficulty in landing tend to focus too much of their attention inside the flightdeck. This can lead to a pilot trying to chase either the Glide Slope or the Flight Director at low level which is damaging to a stable approach. In addition, changing the focus from the electronic display of the PFD/ND and adjusting to the dynamic external visual cues of the runway, aiming point, centre-line tracking, PAPIs and peripheral vision is difficult for an in-experienced pilot. The pilot must look out of the aircraft at the touchdown point on the runway more frequently the nearer he gets to the runway. Once the runway is in sight a good rule of thumb is:

- 1,000’ Look out 25% of the time.
- 500’ Look out 50% of the time
- 250’ Look out 75% of the time
- 100’ Look out 100% of the time

When looking into the flightdeck the student should maintain a radial scan centered on the Attitude and referring to the Airspeed, Vertical Speed and Thrust.

However, once through 500’, this scan reduces to Speed, V/S and N1. At this stage the majority of time is spent looking out.

When thrust or control inputs are required to correct a departure from the speed or path profile remember that once the input has done its job i.e., back on profile, then adjust the thrust or pitch to maintain. The average N1 on approach is 57% for Flaps 30 and 63% for Flaps 40.

Rest the thumb lightly on the autothrottle disconnect switch when guarding the thrust levers on approach. This avoids inadvertent selection of TOGA when autothrottle disconnect is desired.

Pilots are advised to disconnect the A/T first. Having stabilized the thrust manually and with the N1 at a value appropriate to the weight, speed and ROD of the aircraft, the A/P should be disconnected. A deliberate separation between disconnecting the A/T and the A/P will assist in avoiding inadvertent TOGA selections and subsequent goarounds. Only small adjustments to pitch and thrust should be made from then on. Even if the runway is in sight and a decision to land has been made, the PF must always respond to the aircraft call, i.e., 1000 feet, plus 100 etc. as these are also incapacitation calls.

When the threshold passes under the airplane nose and out of sight, shift the visual sighting point to the far end of the runway. Shifting the visual sighting point assists in controlling the pitch attitude during the flare. Maintaining a constant airspeed and descent rate assists in determining the flare point. Initiate the flare when the main gear is approximately 20 feet above the runway by increasing pitch attitude approximately 2° - 3°. This slows the rate of descent.

After the flare is initiated, smoothly retard the thrust levers to idle, and make small pitch attitude adjustments to maintain the desired descent rate to the runway. Ideally, main gear touchdown should occur simultaneously with thrust levers reaching idle. A smooth thrust reduction to idle also assists in controlling the natural nose-down pitch change associated with thrust reduction. Hold sufficient back pressure on the control column to keep the pitch attitude constant.

The concept is:

CHECK (attitude) **CLOSE** (thrust) **HOLD** (attitude)

Do not pump the controls during the flare as this can lead to a difficult landing.

Do not trim during the flare or after touchdown. Trimming in the flare increases the possibility of a tailstrike during touchdown.

Once the main wheels are on the runway, gently lower the nose and select reverse thrust as required.

A go-around can be initiated at any time up to selecting reverse thrust. Once reverse thrust has been selected, you are committed to a landing.

Refer to the FCTM for the bounced landing recovery.

Considerations during landing should include safety issues, i.e. braking action and runway length required/available (making allowance for anticipated runway turnoff point not necessarily the whole length); then should take into account operational considerations such as brake cooling period and schedule constraints; and should focus on saving noise/fuel with the possible use of low drag and idle reverse techniques. The objective should be that the passengers experience a smooth approach, landing and rollout while at the same time maintaining the highest safety and efficiency standards. Smooth application and deselection of brakes or thrust should be normal.

Landing airmanship includes basic pilot technique such as smooth derotation and smooth deselection of autobrake by squeezing rather than 'kicking'. Wet or slippery runway turnoffs present hazards of directional control and reverse thrust should be maintained at idle until achieving taxi speed correct for the conditions. (note: this could be as low as 5 knots for 90-degree turnoff in slippery conditions).

10.3. CROSSWIND LANDINGS

In strong crosswind or turbulent conditions, it may be necessary to alter the landing technique previously described in this chapter. In Virtual Ryanair we use the "de-crab during flare" and the "touchdown with crab technique." Refer to FCTM chapter 6 for crosswind landings.

De-crab during flare:

The objective is to maintain wings level during approach, flare and touchdown.

During the approach, a crab angle is established with wings level in order to maintain the desired track. During the flare, downwind rudder is applied to eliminate the crab and align the aircraft with the centerline. At the same time apply upwind aileron in order to maintain wings level. These cross controls are maintained throughout the landing phase and the control column gradually centralized during the landing roll.

Touchdown with crab:

It is recommended to use this method when landing on slippery runways as it reduces drift on touchdown and allows for rapid deployment of the spoilers and autobrake as all main gear have touched down simultaneously. However, rudder and aileron input to de-crab after touchdown must be applied in order to maintain proper directional control.

This method is not recommended on dry runways with strong crosswind conditions as on landing the aircraft will tend to track upwind until the correct de-crab technique is accomplished. This lack of initial directional control is undesirable. Fly the nosewheel onto the runway after the aircraft is tacking the runway centerline.

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11. POST FLIGHT PROCEDURES

11.1. TAXI IN PROCEDURE

When clear of the active runway and taxi instructions have been received, acknowledged, written in FMC scratchpad and understood by both pilots, the first officer accomplishes the following after landing checks flow:

Transponder ..... ALT OFF

Flaps ..... UP

Trim ..... 5-6 units Nose Up

Autobrake ..... OFF

MFD SYS..... Check hydraulic and brake pressure

Standby ADI ..... OFF

APU (if required) ..... START

Delay until entering parking area.

MCP ..... Set

Radar – OFF

Capt FD – OFF

IAS – 100

Altitude - ODD number plus 100

F/O FD – OFF

ENGINE START switches ..... As required

OFF, unless engine anti-ice is required to parking.

PROBE HEAT switches ..... OFF

Engine cool down recommendations:

- run the engines for at least 3 minutes
- use a thrust setting normally used for taxi operations
- routine cool down times of less than 3 minutes are not recommended.

APU GENERATOR switches (on captain's command) ..... ON

11.2. SHUTDOWN PROCEDURE

After the airplane has come to a complete stop, perform the following actions:

Parking brake ..... Set

Parking brake warning light – Illuminated

Electrical ..... On___

Verify APU powering busses. If APU is not to be used, connect external power.

Start levers ..... CUTOFF

FASTEN BELTS switch ..... OFF

PA (captain) ..... DISARM SLIDES AND OPEN DOORS"

ANTI COLLISION light switch ..... OFF

At 20% N2 place the ANTI-COLLISION light switch OFF.

FUEL PUMP switches ..... OFF

CAUTION: Center tank fuel pump switches should be positioned ON only if the fuel quantity in the center tank exceeds 453 kgs.

GALLEY power switch (as installed) ..... As required

WINDOW HEAT switches ..... OFF

WING and ENGINE ANTI–ICE switches ..... OFF

ELECTRIC HYDRAULIC PUMP switches ..... OFF

CAUTION: To avoid unwanted control column movement during gusty tailwind conditions, leave the B electric hydraulic pump switch ON until completing the Secure procedure.

RECIRCULATION FAN switches ..... AUTO

Air conditioning PACK switches ..... As required

APU fuel flow is reduced with both pack switches off.

ISOLATION VALVE switch – OPEN

Engine BLEED air switches ..... ON

APU BLEED air switch ..... OFF

Exterior lights ..... As required

AUTO BRAKE select switch ..... OFF

Flight deck lights ..... As desired

SPEED BRAKE lever ..... DOWN detent

Parking brake ..... SET

Transponder mode selector ..... 2000 then stby

CVR CB ..... IN/OUT (as required)

Cabin door ..... Unlock

The PF calls "SHUTDOWN CHECKLIST."

The PM accomplishes the SHUTDOWN checklist.

11.3. TRANSIT SHUTDOWN PROCEDURE

After the airplane has come to a complete stop, perform the following actions:

Parking brake ..... Set

Parking brake warning light – Illuminated

Electrical ..... On___

Verify APU powering busses. If APU is not to be used, connect external power.

Start levers ..... CUTOFF

FASTEN BELTS switch ..... OFF

NO SMOKING switch ..... OFF

PA (captain ..... DISARM SLIDES AND OPEN DOORS"

ANTI COLLISION light switch ..... OFF

At 20% N2 place the ANTI-COLLISION light switch OFF.

WING and ENGINE ANTI–ICE switches ..... OFF

VOICE RECORDER switch ..... ON

Verify that this switch is on, including during all turnarounds, to record all clearances, briefings and checklists.

Air conditioning PACK switches ..... As required

APU fuel flow is reduced with both pack switches off.

ISOLATION VALVE switch – OPEN

APU BLEED air switch ..... OFF

Exterior lights ..... As required

Transponder mode selector ..... 2000 then stby

CVR CB ..... IN/OUT (as required)

Cockpit door ..... Unlock

The PF calls "TRANSIT SHUTDOWN CHECKLIST."

The PM accomplishes the TRANSIT SHUTDOWN checklist.

11.4. APU USE

Our policy is minimum use of the APU, however:

If ground power is not clearly available with ground crew standing next to it, then we should start the APU

If you are late, you should always start the APU in both of the above, you should shut down the APU as soon as you get ground power connected (Unless required for air conditioning - <5 degrees or >25 degrees.)